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Machine learning the performance of hedge fund

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ABSTRACT

This study utilizes generative AI to predict and classify the performance of hedge funds based on groups of fund characteristics. Compared to commonly used machine learning methods, our method can successfully distinguish high- and low-performing funds across various investment strategies, with the return spread being the highest in the equity hedge strategy at 3.16 % monthly. The results are robust in risk-adjusted return prediction. Trend-based features are the most important predictors of future fund performance. Returns of predictive long-short portfolios are higher following periods of low narrative attention and favorable macroeconomic conditions. The asset allocation exercise highlights the significant economic value of machine learning. Our study enriches the burgeoning field of machine learning and artificial intelligence for finance by applying big data techniques to fund selection and allocation.

1. Introduction

Over the last 20 years, the hedge fund industry has undergone substantial expansion, with the total size of hedge funds reaching unprecedented levels.[1] Hedge fund managers are generally regarded as sophisticated and more willing to take risks, utilizing dynamic strategies and displaying time-varying market exposure (Fung, Hsieh, Naik, and Ramadorai, 2008; Cao, Chen, Liang, and Lo, 2013). Hedge funds' performance, compared to mutual funds, is more variable and reflects the expertise of these talented managers. With the rapid growth in the application of machine learning in the financial market, a natural question is whether machines can learn the performance of hedge funds and identify which, if any, characteristics can help separate hedge fund performance. The use of machine learning in hedge funds can help us uncover the complex interactions among different features and discover new market anomalies, thus providing a basis for developing new investment decisions for hedge fund managers.

Beyond the commonly used machine learning models (Panopoulou and Vrontos, 2015; Slavutskaya, 2013) for predicting hedge fund returns, this paper introduces a variant of the popular generative transformer model that has emerged in recent years, known as the temporal fusion transformer (TFT) model. The TFT model is particularly advantageous for predicting returns due to its ability to handle diverse data types and complex time series data. It can integrate various financial indicators, macroeconomic variables, and hedge fund-specific characteristics, leading to more accurate predictions. For example, considering the Temporal dependency processing module in TFT, it contains a sequence-to-sequence layer which can capture short-term temporal relationships within the input

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[1] In 2022, the total size of hedge funds in the United States is estimated to be around $3.8 trillion.

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sequence, allowing the model to understand patterns and trends over time and a multi-head attention module which can focus on information from different positions in the input sequence, making the model more effective in handling long-distance dependencies.

We select hedge funds that are actively trading from 1994 to 2018 from the Lipper TASS database. The target variable we are predicting is hedge fund return, including raw return and abnormal return. The abnormal return is defined as the Fung and Hsieh (2004) seven-factor alpha of the fund. Meanwhile, to avoid look-ahead bias, we use lagged predictors with one quarter to predict fund return. Our predictors consist of 17 fund and fund-related characteristics, including 7 trend-based characteristics, 7 risk-based characteristics, and 3 macro-derivative features. Moreover, we introduce hedge fund-related economic narrative provided by Hong, Jiang, Meng, and Xue (2024) to explore the model performance in high and low media attention. The hedge fund-related topic is estimated using the Latent Dirichlet Allocation (LDA) model proposed by Blei, Ng, and Jordan (2003) from The *Wall Street Journal* and the vocabulary includes terms such as 'fund', 'investor', 'asset', 'investment', 'money', 'manager', 'invest', 'return', 'hedge' and 'portfolio'.

The empirical finding is noting. First, we document the significant predictability of the TFT model in hedge fund samples. Considering the equal-weighted (predicted return-weighted) portfolios sorted by predicting results, the highest decile generates monthly average return of 0.52 % (0.59 %) with t-stat 4.28(4.37) compared to the return of −0.08 % (0.02 %) with t-stat of −0.76 (0.20) in the lowest decile. By comparing with traditional linear and nonlinear models, the TFT model not only outperforms traditional models in terms of the magnitude of returns obtained, but also excels in the stability of returns. The highest decile of the TFT model can achieve a minimum drawdown of 8.19 %. Considering transaction costs, the best decile of hedge funds still achieves a monthly average return of 0.32 % (0.39 %) with t-stat of 2.69(3.15).

We then examine the robustness of machine learning and introduce TFT model to nine different types of hedge funds. The return spread among all strategies is statistically significant. Specifically, the top decile in Fixed Income Arbitrage strategy generates the highest monthly average return of 0.76 % with t-stat of 6.54. The bottom decile of Fund of Funds strategy makes the lowest monthly average return of −0.22 % with t-stat of −2.14.

Second, we explore the characteristic importance in prediction. We separately use different characteristic theme to predict hedge fund return and report the predicting accuracy. We then refer to the method of Breiman (2001) and used permutation feature importance method to study the importance of individual characteristic. The result shows that trend-based characteristic theme is more important than other themes. The total importance of trend-based features is 59.35 % (Sum up the importance of all individual features under this panel). The importance of risk-based features is 21.47 %, as the second important theme. The most important individual characteristic is 6-month return with importance value of 14.69 %.

Third, as a large set of hedge funds has struggled to achieve significant abnormal returns post-2008, we calculate the alphas of our decile portfolio and find hedge funds in the highest decile can generate significant alpha of 0.42 % with t-stat of 3.50. We then explore the source of these abnormal returns from cross-sectional and time-series perspectives. We first regress the decile alpha on several fund-related features and find that factors such as the fund age, leverage and incentive fees significantly contribute to the fund alphas. Additionally, we find that the redemption period significantly affects the alpha value of hedge funds only in the highest decile portfolio. Second, from time-series perspective, we divide the sample with high and low media attention and find that the significant abnormal returns in the highest and lowest deciles are primarily driven by hedge fund managers' active choices during periods of low attention period. Furthermore, when categorizing the sample based on seven macroeconomic variables, we find that abnormal returns are higher during economic upswings.

Finally, we evaluate the economic gains of the TFT model by considering a mean–variance investor who reallocates wealth monthly between hedge funds and a risk-free asset based on machine learning forecasting results. The results show that this strategy generates a positive certainty equivalent return (CER). Among the portfolios, the Fixed Income Arbitrage strategy performs the best, achieving a CER of 0.52 % and a Sharpe ratio of 0.46. Additionally, we analyze an active investment strategy that combines a passive portfolio (60 % stocks and 40 % bonds) with hedge funds. Our findings indicate that as the allocation to hedge funds increases, investors experience positive and growing utility.

Our study makes three main contributions. First, this paper provides new economic evidence on timing the hedge fund via generative AI. Previous studies document that machines can learn the skill of mutual fund managers and generate notable profit (Ron, Zihan, Markus, and Stijn, 2023; DeMiguel, Gil-Bazo, Nogales, and Santos, 2023). Wu, Chen, Yang, and Tindall (2021) compare the machine learning approach for equity hedge fund return prediction with the price trend-related characteristics and macro features. Gong, Guan, and Chen (2022) combine textual language with machine learning methods to predict crude oil futures pricing. Shu and Tiwari (2022) demonstrate that traditional machine learning models can achieve better performance than the Fung and Hsieh (2004) model. While machine learning applications are becoming increasingly widespread, there is limited literature exploring its use in hedge funds. This paper first proposes a generative AI approach, the TFT model, to investigate this domain. TFT model is a novel and powerful deep learning method developed by Lim, Arık, Loeff, and Pfister (2021) and has recently become popular as the foundation model (such as ChatGPT) in dealing with sequential data. In contrast to various other machine learning models (Gu, Kelly, and Xiu, 2021; Giglio, Kelly, and Xiu, 2022; Ron et al., 2023; Leippold, Wang, and Zhou, 2022), the self-attention-based mechanism employed in this generative AI facilitates the interaction and learning of dominant features among the inputs, while the resulting outputs aggregate these interactions and attention scores.

Second, this paper echoes several studies on hedge funds and highlights the characteristic of its performance. In previous research, DeMiguel et al. (2023) find that historical performance is the most important predictor of mutual fund performance. Kosowski, Naik, and Teo (2007) argue that hedge funds achieve significant abnormal returns compared to mutual funds. Gregoriou and Rouah (2002), Howell (2001), and Amenc, Martellini, and de Chardin (2003) demonstrate a significant correlation between hedge fund age, management fees, and incentive fees with abnormal returns. This paper extends these findings using big data and new measurement

methods.

Third, we enhance the understanding of hedge fund with economic narratives. A growing body of literature discusses the role of news media in spreading narratives (Bybee, Kelly, Manela, and Xiu, 2021; Levy, 2021) and driving economic expectation (Coibion, Gorodnichenko, and Weber, 2022). Hong et al. (2024) construct a Narrative attention index using news topics extracted from the *Wall Street Journal* and find the predictability of Macro status. In this paper, we follow Hong et al. (2024) and select hedge fund-related topics to identify periods of high and low narrative attention/sentiment, and find the different performance of hedge funds during these periods.

The rest of this paper is organized as follows. Section 2 discusses the related literature. Section 3 describes the data and methods, Section 4 and Section 5 present the empirical findings and economic ground, and Section 6 summarizes the paper.

2. Related literature

The previous literature examines the risk-return characteristics of hedge funds. Titman and Tiu (2011) regress the hedge fund returns on a series of risk factors and find the low R-squares of returns on factors generate higher performance, the funds with higher information ratios charge higher incentive and management fees. Bali, Brown, and Caglayan (2012) separate the total risk into systematic and residual components and find the systematic part is the significant factor in explaining returns. Cao et al. (2013) show the fund manager can time the hedge fund with liquidity risk. Patton and Ramadorai (2013) examine the fund exposure to risk factors using high-frequency data. Our paper revisits the linkage between characteristics with hedge fund returns from a high - dimensional perspective.

In the age of big data, machine learning-based models have garnered significant attention in finance due to their strong predictive performance. Previous studies have emphasized the advantages of machine learning methods in measuring asset risk premiums, such as the use of deep learning models to estimate the stochastic discount factor (Chen, Pelger, and Zhu, 2019), the superior performance of linear and nonlinear models for empirical asset pricing (Gu, Kelly, and Xiu, 2020; Ma, Wang, and Chen, 2023), the more accurate stock return predictions based on image analysis methods (Jiang, Kelly, and Xiu, 2023), and the introduction of an autoencoder asset pricing model with smaller out-of-sample pricing errors compared to former factor models (Gu et al., 2021). However, few studies have examined the use of machine learning methods for fund performance. Recently, Kaniel, Lin, Pelger, and Van Nieuwerburgh (2023) explored several machine learning approaches to predict mutual fund performance. The similar research is shown in DeMiguel et al. (2023).

Regarding interpretability in machine learning, Gu et al. (2020) use feature importance to analyze the key factors affecting asset prices, and Molnar (2020) construct permutation feature importance based on a data selection approach to further elucidate the relationship between features and returns. Kelly, Pruitt, and Su (2019) study the linear contribution of stock features in constructing the risk premium (loading) and obtain a ranking of feature importance. Kapetanios and Kempf (2022) use deep learning to adapt the traditional Fama-Macbeth approach and construct an interpretable neural network model. Fan, Ke, Liao, and Neuhierl (2022) develop a structural deep learning model and provide an economic interpretation by decomposing the neural network's predictions into risk-related and mispricing components. The authors then utilize deep learning to forecast long- and short-term expected returns and further decomposed the model's forecast source. Our paper extends the discussion of interpretable machine learning and provides the empirical evidence.

3. Data and methodology

3.1. Data

We select a sample of hedge funds from the Lipper TASS database, a widely used dataset in hedge fund research. Our sample period spans from January 1994 to December 2018[2] and comprises nine fund categories: convertible arbitrage, dedicated short bias, emerging markets, event driven, fix income arbitrage, long/short equity hedge, equity market neutral, global macro, and multi-strategy. In total, our data contain over 5000 funds to train the model effectively.

Our predictors comprise three groups. The first group is trend-based characteristics, including 1-month to 12-month cumulative returns (R1, R3, R6, R9, R12), Upside Potential (UP), and First-order autocorrelations of returns (ACF1). The second group, risk-based features, consists of VAR/ES at empirical distribution and Cornish-Fisher distribution (NPVAR, NPES, CFVAR, CFES), Distinctiveness of the strategy (SDI), Skewness and Kurtosis of returns (Skew and Kurt). The last component is related to Macro-derivative features, which are Liquidity Beta (Illiqbeta), Economic uncertainty beta (Euncbeta) and Volatility beta (Vixbeta).[3] We have also added two panels to further discuss the characteristics of hedge funds. The fundamental characteristics of hedge funds consist of a group, including the size and age of hedge funds (Size and Age), management fees, incentive fees, redemption period, minimum investment amount, lock-up period, and leverage ratio of hedge funds (Manfee, Incfee, Redem, Mininvest, D_lockup, D_leverage). Moreover, Hong et al. (2024) make use of an extensive news database sourced from The *Wall Street Journal*, which includes more than 880,000 articles spanning the years 1984 to 2021. They apply topic modeling methods from the domain of natural language processing (NLP) to

[2] Although the database contains fund returns back to November 1977, it does not retain dead funds until 1994 and data from the early period contain survivorship bias.

[3] We report the description and construction of each characteristic in appendix with OA1.

Table 1
Descriptive statistics of corporate bond factors.

	Definition	Mean	Std	Min	25 %	75 %	Max
Panel A: Trend − based Characteristics							
R1	1 − month return	0.48	1.76	−4.73	−0.44	1.45	5.42
R3	3 − month accumulative return	1.47	3.36	−13.60	−0.45	3.42	15.90
R6	6 − month accumulative return	3.02	5.09	−23.20	−0.07	6.05	30.05
R9	9 − month accumulative return	4.63	6.54	−30.19	0.67	8.51	37.80
R12	12 − month accumulative return	6.30	7.81	−36.21	1.50	10.96	45.05
ACF1	Lag 1 auto correlation of the past 36 months' return	0.14	0.19	−1.00	0.06	0.23	0.99
Upside Potential	The max return in past 36 months	5.57	3.49	1.48	3.11	5.67	17.15
Panel B: Risk − based Characteristics							
NPVAR	Value at Risk at empirical distribution	−3.03	2.05	−9.53	−3.40	−1.69	0.09
CFVAR	Value at Risk at Cornish − Fisher distribution	−2.94	2.10	−9.87	−3.13	−1.58	0.10
NPES	Expected Shortfall at empirical distribution	−4.56	3.08	−14.72	−4.97	−2.47	−0.19
CFES	Expected Shortfall at Cornish − Fisher distribution	−4.40	2.72	−13.85	−4.40	−2.67	−0.43
SDI	Distinctiveness of fund − investment strategy	0.34	0.30	−0.99	0.15	0.55	1.00
Skew	Skewness of the past 12 months' returns	−0.18	0.64	−1.39	−0.63	0.27	1.02
Kurt	Kurtosis of the past 12 months' returns	0.12	1.29	−1.47	−0.85	0.77	3.34
Panel C: Macro − derivative Features							
Liquidity Beta	Loading on market liquidity	2.75	5.77	−10.71	−0.02	4.95	17.58
Economic − uncertainty Beta	Loading on economic uncertainty	−0.04	1.08	−2.88	−0.35	0.25	2.76
Volatility Beta	Loading on VIX index	0.03	0.12	−0.26	−0.03	0.07	0.35
Panel D: Fundamental Features							
Size	Nav	39.00	379.78	0.86	1.26	13.95	21001.16
Age	Age	70.37	49.13	0.00	34.00	93.00	492.00
Manfee	Managementfee	1.39	0.59	0.00	1.00	1.75	10.00
Infee	Incentivefee	12.71	8.01	0.00	7.50	20.00	50.00
Redem	Redemption notice period	34.74	31.02	0.00	5.00	45.00	365.00
Mininvest	Minimum investment	5.22	133.56	0.00	0.05	1.00	5000.00
D_lockup	Lockup period	2.23	5.83	0.00	0.00	0.00	90.00
D_leverage	Style leverage	3.84	46.00	−22.21	−0.18	1.54	881.76
Panel E: Narrative − based Feature							
NFA	Narrative fund attention	1.11	0.40	0.51	0.81	1.39	2.71

The opening row of the table presents the mean monthly return, standard deviation, and additional statistics of hedge fund returns. The remaining rows provide statistical information on the predictors. The construction of each predictor is further elaborated in the Appendix. The dataset is derived from the Lipper TASS database and spans the period from January 1994 to December 2018. We report the construction of characteristic in Appendix A.

aggregate and break down economic narratives into quantifiable and temporally variable news topics. The time series analysis of 180 different news topics encapsulates the media's focus on each subject. Based on this extensive dataset, our paper selects news topics related to 'fund'. Subsequently, we calculate the average of this metric, named narrative fund attention. Fund topics include 'fund', 'investor', 'asset', 'investment', 'money', 'manager', 'invest', 'return', 'hedge' and 'portfolio'. The sample period for attention ranges from January 1984 to June 2021. Table 1 reports the descriptive statistics of the data. Considering the difference in the order of magnitude between different variables, the monthly cross-sectional data of each predictor is normalized.

3.2. *Temporal fusion transformer model*

Temporal fusion Transformer (TFT) model[4], a type of attention-based deep learning model, is adopted in this study to predict hedge funds returns. TFT model mainly consists of four modules, (1) Gating module can flexibly handle nonlinear features where needed. It primarily consists of two components: the ELU (Exponential Linear Unit) activation function and the GLU (Gated Linear Unit), which can suppress any unnecessary parts of the architecture for specific datasets. (2) Variable selection network can learn using only the most significant features, greatly improving the model's performance. (3) Static covariate encoder integrates static features into the network, adjusting temporal dynamics through contextual vector encoding. (4) Temporal dependency processing module contains a sequence-to-sequence layer which can capture short-term temporal relationships within the input sequence, allowing the model to understand patterns and trends over time and a multi-head attention module which can focus on information from different positions in the input sequence, making the model more effective in handling long-distance dependencies. The specific implementation of the

[4] We tabulate the details and hyperparameters setting of our models in appendix.

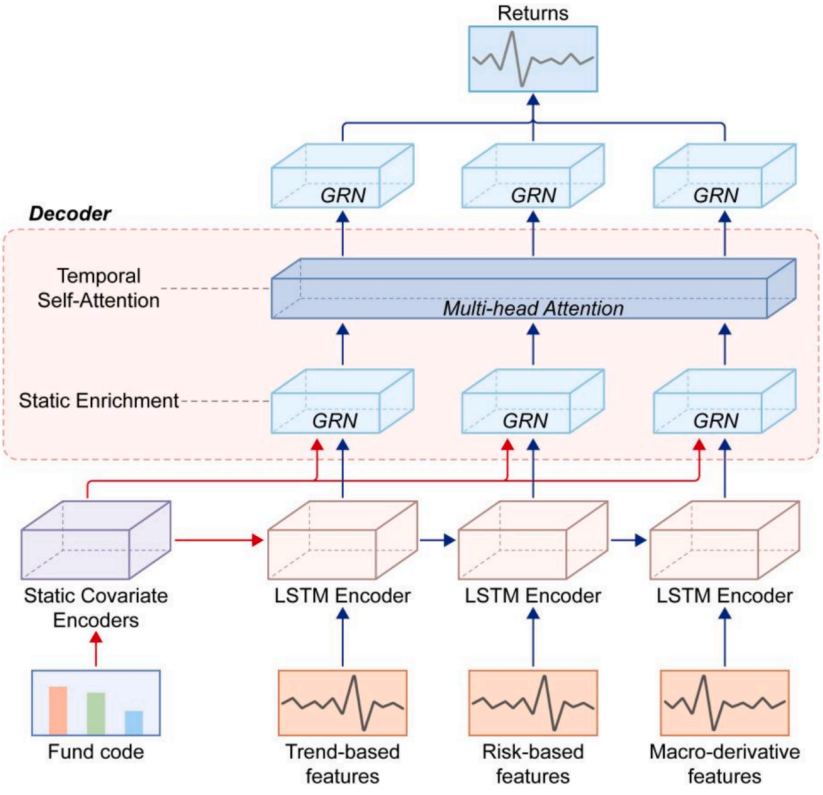


Fig. 1. Temporal Fusion Transformer model. This figure shows the internal structure of our constructed TFT model, which consists of 1 embedding layer, several encoder and hidden layers, and 1 output layer. where the input data are n features of P funds. The Table A1 in online appendix shows hyper parameter setting of the model.

modules mentioned above will be presented in the Appendix. The overall construction is depicted in Fig. 1 and simplified function (1) is listed below.

$$R_{i,t} = f_t\left(Z_{i,t-4}\right) + \mu_{i,t}, \tag{1}$$

$f_t\left(Z_{i,t-4}\right)$ represents nonlinear function obtained through our machine learning algorithms that extract pricing information from the data set $Z_{\{i,t-4\}}$. Due to the lag in obtaining hedge fund data from various channels in reality, we select variables lagged by one quarter as our feature variables. Importantly, we incorporate the 'product of reference' factor into the static input, and the rest of the factors are placed into past-observed inputs. We divide our data into three subsamples to train and test f_t in equation (1) with TFT model. The training period starts from January 1994 to December 2001, the validation period is from January 2002 to December 2006, and the out-of-sample period is from January 2007 to December 2018. In the model's predictions, we employ an expanding window approach, which means that once the model is trained, we do not adjust its parameters and proceed to make predictions on the test set data.

4. Empirical results

4.1. The fund performance sorted by predicted returns

After training the TFT model, we use all 17 features listed in Table 1 to predict the fund's returns. We divide the funds into deciles based on the predicted returns for the next month. Within the deciles we weight the funds either by their predicted value or equally. Under the weighting scheme based on prediction results, we apply a certain transformation to the predicted returns for the deciles that deviate from the center.[5]

[5] The weights based on predictions are defined as the following adjusted and scaled weights:For the highest decile of funds:$\widetilde{\mu}_{i,t} = \widehat{\mu}_{i,t} - \min_{i \in Top} \widehat{\mu}_{i,t}$ (2) For the lowest decile of funds:$\widetilde{\mu}_{i,t} = \widehat{\mu}_{i,t} - \max_{i \in Bottom} \widehat{\mu}_{i,t}$ (3) $w_{i,t}^{pred} = \frac{\widetilde{\mu}_{i,t}}{\sum_{i=1}^{n} \widetilde{\mu}_{i,t}}$ (4)where $\widehat{\mu}_{i,t}$ are the predictions of TFT model. For the highest and lowest deciles of hedge funds, we subtract the minimum and maximum values from their predicted results, respectively. This allows us to assign a higher weight $w_{i,t}^{pred}$ to funds with greater absolute predicted values.

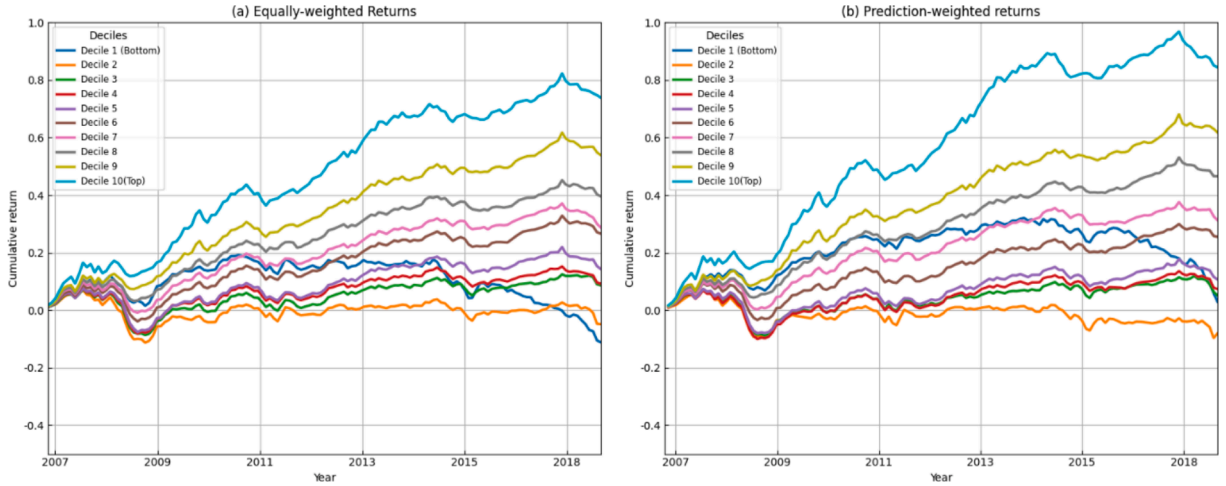


Fig. 2. Cumulative returns. These figures show the cumulative returns sorted into prediction deciles when considering the complete information set (Trend-based, Risk-based and Macro-derivative characteristics) to predict returns.

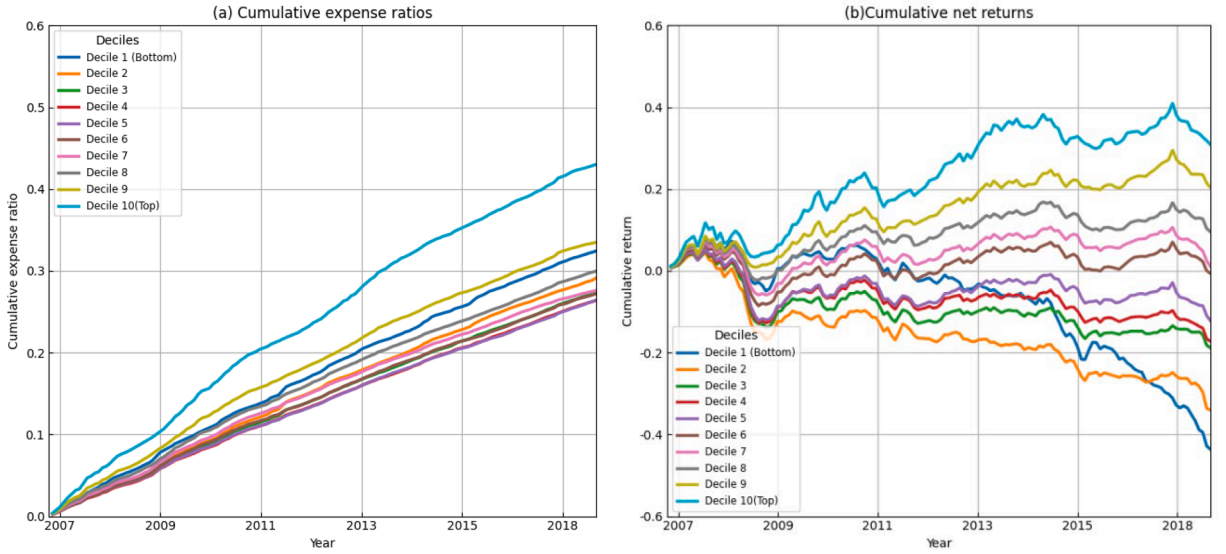


Fig. 3. Transaction costs and cumulative returns after transaction costs. The left figure shows the cumulative expense ratios of equal-weighted deciles based on the full information set (Trend-based, Risk-based and Macro-derivative characteristics). The right figure shows the net returns for the equal-weighted deciles, that is, the returns minus the fees.

Fig. 2 illustrates the cumulative returns from investing in the decile portfolios of hedge funds. Panel (a) of Fig. 2 adopts an equal weighting approach, assigning identical weight to each fund within a decile. In contrast, Panel (b) employs portfolio weights derived from the TFT model's predictions, termed 'prediction-weighted returns'. The findings reveal a cumulative return of 73.88 % for equal weighting and 84.44 % for prediction weighting when investing in the highest decile of funds. This indicates that the TFT model can not only predict the relative ranking of hedge fund returns, but also accurately capture the specific values of hedge fund returns. Although allocating funds based on predictive weights can achieve a higher cumulative return than allocating funds based on equal weights, in order to reduce the complexity of trading, we will use the equal-weighted method for fund allocation in the subsequent analysis.

In addition, according to the predictive results of the TFT model, when investing in the lowest decile of hedge funds, the equal weight method and the prediction-based weight method can respectively achieve cumulative returns of −11.19 % and 2.98 %. This indicates that while the TFT model can help us select hedge funds with outstanding performance, it has disadvantages in terms of risk avoidance.

Finally, Kaniel et al. (2023) apply neural network models to mutual funds and back test the cumulative returns of mutual funds based on the predictions of the neural network model from 1980 to 2020. The cumulative return of the highest decile portfolio is only

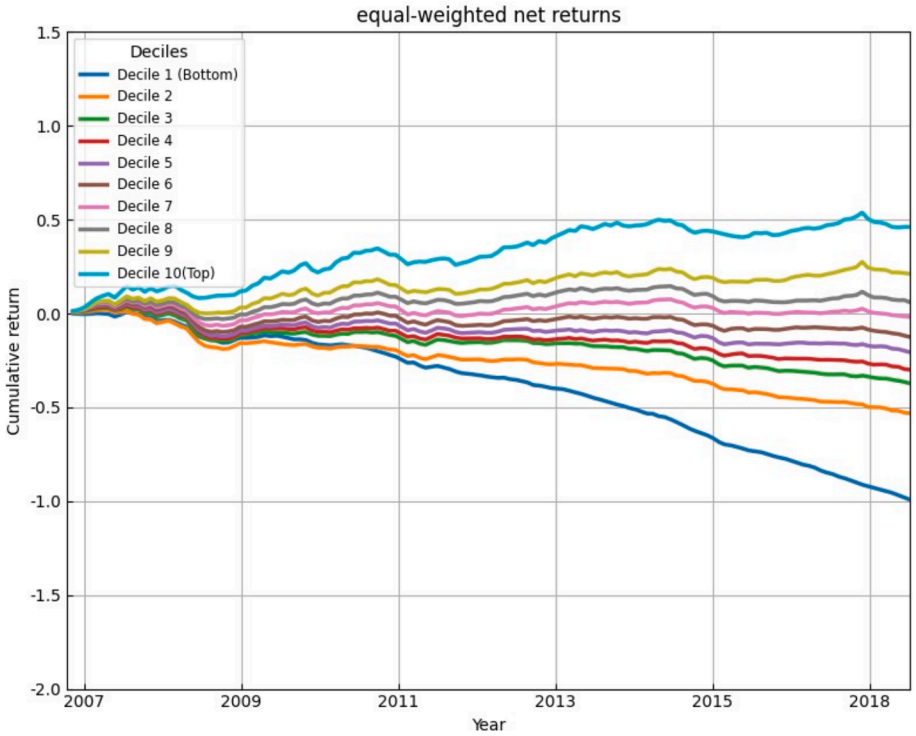


Fig. 4. Cumulative returns based on after-fee performance predictions. This figure shows the cumulative after-fee returns for prediction-sorted decile portfolios. We use fund-specific characteristics to predict after-fee returns and calculate the equal-weighted cumulative return.

72 %, while this cumulative return is 73.88 % when using the TFT model's predictions to classify hedge funds from 2007 to 2018. On the one hand, this is because hedge funds can obtain higher abnormal returns than mutual funds, and on the other hand, it is because the TFT model we use is more advanced and has more advantages in financial markets. Interestingly, deep learning seems to demonstrate better predictive power for underperforming funds in mutual funds, while showing better predictive power for outperforming funds in hedge funds.

Next, we incorporate transaction costs into our analysis of the predicted decile portfolios. Mutual funds and hedge funds have different fee structures. Mutual funds generally charge a lower proportion of management fees, and the larger the size of mutual funds, the higher the management fees charged. Unlike mutual funds, hedge funds usually adopt the fee structure of management fees plus incentive fees, and hedge funds mainly rely on the profit sharing of customers to earn income, so the extraction ratio of incentive fees is much higher than that of management fees, usually adopting the '2% plus 20 %' model. In addition, in order to avoid the problem of investors paying more due to the drawdown of hedge funds, there are usually high-water clauses and recovery clauses when investing in hedge funds.

Panel (a) of Fig. 3 displays the cumulative transaction costs of the decile portfolios. According to the research results of Kaniel et al. (2023), there is no significant difference in the cumulative transaction costs of the highest decile and lowest decile of mutual funds. However, we find there is a significant difference in cumulative fees in hedge funds. According to Fig. 3, the highest decile of hedge funds has the highest transaction costs with 43.01 %. Different fee structures lead to differences in cumulative transaction costs between hedge funds and mutual funds. The management fee plus incentive fee structure of hedge funds results in the highest transaction costs for high-performing hedge funds. However, it is noteworthy that the transaction costs of the worst-performing hedge funds are only second to those of the second-best-performing hedge funds, suggesting that they seem to charge higher management fees.

Panel (b) of Fig. 3 shows the cumulative returns after deducting the fees of hedge funds. Even after considering transaction costs, the highest decile of funds still manages to achieve around 30.87 % in cumulative positive returns, making them the group with the highest net returns among the ten categories. This demonstrates that our return predictions can indeed assist investors in realizing practical returns.

To assess the practical application of our predictive analysis, Fig. 4 showcases the cumulative returns achieved by adopting an equal-weighted investment strategy, where returns, post-transaction cost deductions, are treated as the predicted values. This approach highlights the effectiveness of incorporating transaction costs into our predictive model to gauge the actual investment outcomes. Notably, the highest decile of hedge funds is able to achieve a cumulative return of 46.10 % when returns net of transaction costs are used as predicted values, which is higher than the 30.87 % post-fee cumulative returns we obtained when using raw returns as predicted values. Furthermore, the analysis reveals that four fund groups record cumulative returns above zero, underscoring the TFT model's utility in a real-world investment scenario. This evidence bolsters our confidence in the TFT model's capacity to guide

Table 2
The performance of hedge funds using different models.

Model	Portfolio	α(%)	IR	Mean(%)	Std(%)	SR	MaxDD(%)	MPPM(%)
OLS	Low	−0.21	−0.21	−0.13	1.12	−0.12	24.71	−1.96
	2	0.01	0.01	0.09	1.03	0.08	11.39	0.73
	3	0.04	0.04	0.11	0.97	0.12	9.20	1.09
	4	0.06	0.07	0.14	0.97	0.15	9.54	1.44
	5	0.09	0.10	0.17	1.00	0.17	9.29	1.72
	6	0.08	0.09	0.16	1.02	0.16	11.17	1.65
	7	0.11	0.12	0.20	1.04	0.19	11.12	2.01
	8	0.10	0.10	0.19	1.11	0.17	11.39	1.87
	9	0.16	0.15	0.25	1.15	0.21	11.71	2.56
	High	0.27	0.21	0.37	1.39	0.26	15.18	3.80
PCA	Low	0.06	0.07	0.15	1.04	0.14	10.81	1.43
	2	0.07	0.07	0.15	1.03	0.14	10.81	1.45
	3	0.07	0.07	0.15	1.06	0.14	10.77	1.47
	4	0.09	0.09	0.17	1.06	0.16	10.42	1.68
	5	0.09	0.09	0.17	1.05	0.16	10.76	1.74
	6	0.06	0.07	0.14	1.06	0.13	10.28	1.38
	7	0.05	0.05	0.14	1.07	0.13	10.65	1.30
	8	0.07	0.07	0.15	1.04	0.15	10.76	1.53
	9	0.08	0.09	0.16	1.05	0.15	10.31	1.60
	High	0.07	0.08	0.16	1.06	0.15	10.31	1.54
IPCA	Low	−0.37	−0.34	−0.26	1.21	−0.21	36.91	−3.55
	2	−0.10	−0.10	−0.01	1.13	−0.01	13.30	−0.46
	3	−0.02	−0.02	0.07	1.07	0.07	12.48	0.51
	4	0.01	0.01	0.10	1.07	0.09	12.11	0.81
	5	0.08	0.09	0.17	1.06	0.16	11.78	1.69
	6	0.08	0.08	0.16	1.05	0.15	11.25	1.54
	7	0.12	0.13	0.20	1.03	0.20	9.92	2.11
	8	0.22	0.23	0.30	1.06	0.28	9.26	3.21
	9	0.28	0.28	0.34	1.09	0.31	10.67	3.72
	High	0.42	0.35	0.48	1.27	0.37	13.63	5.21
AE	Low	−0.08	−0.07	0.03	1.30	0.02	13.26	−0.19
	2	−0.04	−0.04	0.06	1.17	0.05	13.45	0.32
	3	0.00	0.00	0.10	1.09	0.09	12.39	0.84
	4	0.00	0.00	0.09	1.05	0.09	11.41	0.76
	5	0.03	0.03	0.11	1.06	0.10	11.02	1.00
	6	0.07	0.08	0.16	1.04	0.16	10.13	1.62
	7	0.12	0.14	0.20	0.98	0.20	8.97	2.06
	8	0.15	0.17	0.22	0.99	0.22	8.98	2.30
	9	0.21	0.23	0.28	1.03	0.27	9.54	3.04
	High	0.24	0.20	0.30	1.26	0.24	10.10	3.08
TFT	Low	−0.17	−0.15	−0.08	1.23	−0.06	26.63	−1.39
	2	−0.11	−0.11	−0.03	1.11	−0.03	17.00	−0.78
	3	−0.02	−0.02	0.06	1.04	0.06	14.91	0.39
	4	−0.03	−0.03	0.06	1.05	0.06	14.77	0.44
	5	0.01	0.02	0.10	1.08	0.09	14.40	0.85
	6	0.10	0.11	0.19	1.04	0.18	12.15	1.90
	7	0.12	0.13	0.20	1.04	0.19	10.37	2.09
	8	0.20	0.22	0.28	1.01	0.27	7.43	3.01
	9	0.29	0.28	0.38	1.11	0.34	7.61	4.15
	High	0.42	0.31	0.52	1.44	0.36	8.19	5.57

This table shows the performance results of hedge funds under different models. We divide the forecast results of the rate of return into 10 groups from low to high according to each model and calculated the performance of hedge funds in each group. In the performance of hedge funds, Mean is the monthly return of hedge funds, and Std is its standard deviation. SR is the Sharpe ratio, alpha is estimated based on the Fung and Hsieh (2004) seven-factor model, and IR is the information ratio. MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007), estimated using a risk-aversion coefficient of 5, while MaxDD is the maximum drawdown.

investors towards profitable decisions, even after accounting for the often overlooked but significant impact of transaction costs.

In addition, we compare the predictive ability of the TFT model with that of traditional machine learning methods, that is OLS, PCA, IPCA model from Kelly et al. (2019) and AE model from Gu et al. (2021). Based on the predictive results of each model, we construct decile portfolios for hedge funds and calculate the performance of each portfolio during the period of 2007–2018. Since hedge funds do not have a short-selling mechanism, we pay more attention to the performance of the top group. As shown in Table 2,

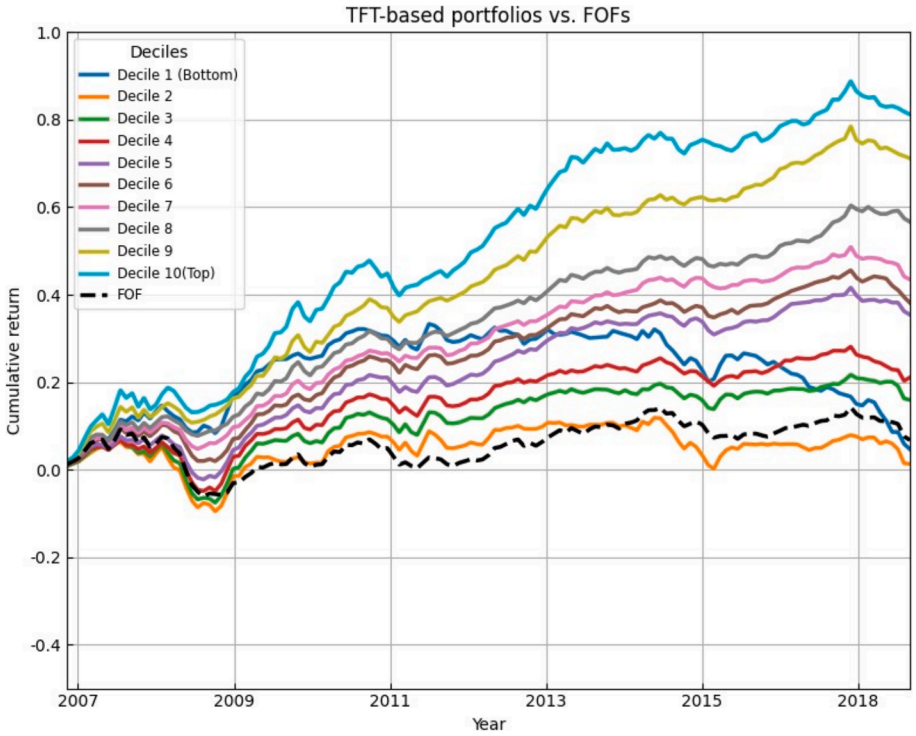


Fig. 5. TFT-based portfolios vs. FOFs. This figure shows a comparison of the cumulative returns from 2007 to 2018 between portfolios constructed based on the prediction results of the TFT model and actual Fund of Funds (FOFs). When constructing the portfolio according to the TFT model, we use the equal-weighting method and exclude hedge funds using the Fund of Fund strategy in advance, representing their cumulative returns with colored lines. In calculating the returns of the actual FOFs, we compute the average monthly returns of all FOFs and represent their cumulative returns with a black dashed line.

the TFT model can achieve the highest average monthly return and abnormal return, which are 0.52 % and 0.42 % respectively, followed by the IPCA model. However, the portfolio constructed according to the TFT model has a relatively high volatility, at 1.44 %. Nevertheless, the maximum drawdown brought by the TFT model to investors is the smallest, and investors face a smaller downside risk. MPPM is the manipulation-proof performance measure of Goetzmann, Ingersoll, Spiegel, and Welch (2007). The TFT model has the best MPPM performance, at 5.57 %. Among all the models, the PCA model performs the worst and has the weakest profit-making ability.

Finally, we compare the performance of actual FOFs with portfolios constructed based on the prediction results from the TFT model. Although there are some FOFs in reality that are not in the database or have already ceased to exist, according to the research of Joenväärä, Kauppila, and Kahra (2021), such comparisons are still meaningful. When constructing the portfolio based on the TFT model, we exclude hedge funds using the FOF strategy to ensure comparability. Although our comparisons are not perfect, they still allow us to see the advantages of TFT-based portfolios relative to FOFs.

Fig. 5 presents a comparison of the cumulative returns between TFT-based portfolios and FOFs. By the end of 2018, the cumulative returns that FOFs are able to achieve are only 6.74 %, which is higher than those of only two TFT-based investment portfolios. Furthermore, after the exclusion of FOFs, the highest decile of the portfolio constructed using the TFT model achieves a higher cumulative return of 81.22 % compared to before the exclusion. This further highlights the significance of the TFT model in hedge fund forecasting.

4.2. Long holding periods

Exploring the impact of holding period duration on portfolio performance, Fig. 6 offers a detailed examination through its four panels. This analysis is crucial for understanding how time influences both the returns and risks associated with long-short investment strategies.

Panel (a) reveals an interesting trend: as investors extend their holding period, the average return of their investment portfolios tends to increase, reaching a peak around 10 to 12 months, after which the average return begins to level off or decline. This seems to suggest that after predicting the returns of hedge funds, we should hold onto the highest decile of hedge funds for the long term, as this can help us achieve higher returns.

Panel (b) addresses the risk factor, illustrating an increase in the standard deviation of returns as holding periods lengthen. This correlation between longer durations and heightened risk challenges investors to balance their desire for higher returns against the

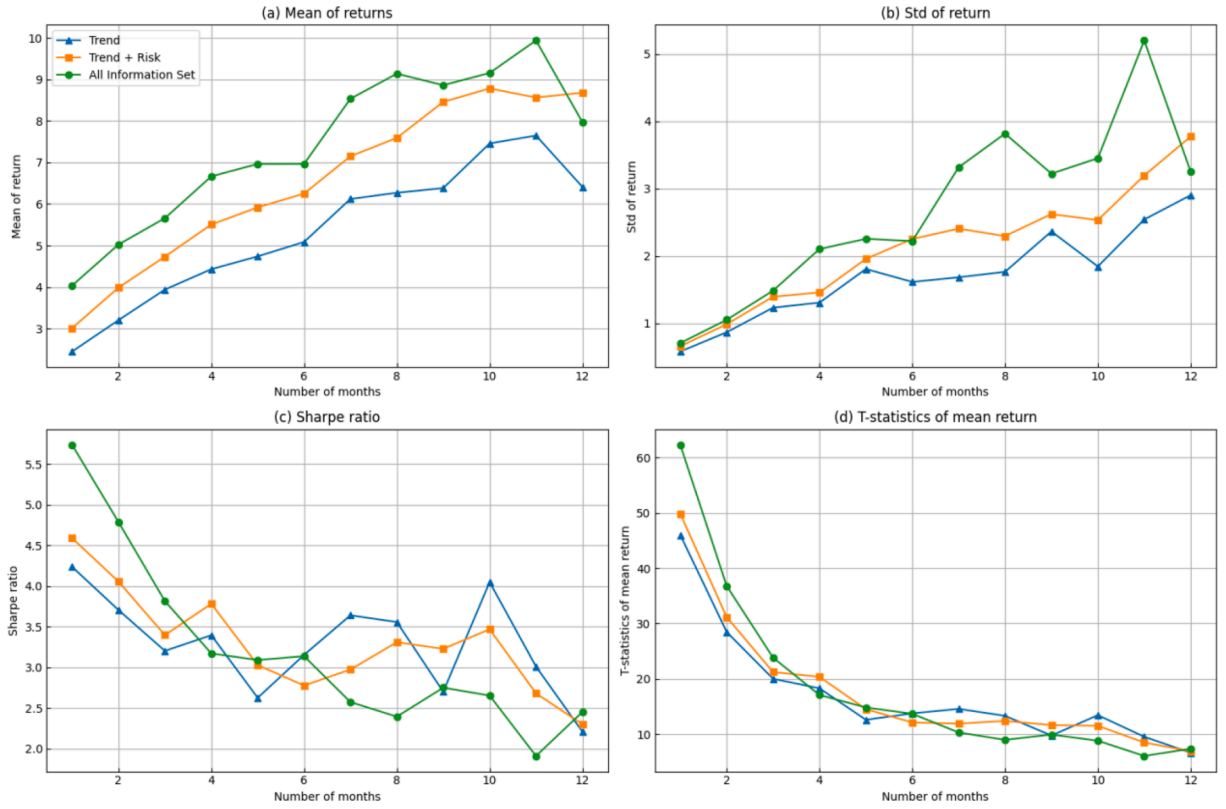


Fig. 6. Longer holding period. These figures show the results for long-short equal-weighted portfolios for different holding periods. At each time t, we sort funds based on the one-month prediction into deciles and hold the long-short prediction portfolio for s months with overlapping returns. We calculate the mean, Sharpe ratio, standard deviation, and t-statistics of the overlapping returns. The one-month prediction uses either Trend-based characteristics, Trend-based + Risk-based characteristics or Trend-based + Risk-based + Macro-derivative characteristics.

increased uncertainty they must shoulder.

The analysis deepens with Panel (c), which plots the Sharpe ratios over the same holding periods. As the holding period increases, the Sharpe ratio shows an initial decline followed by a increase. This indicates that during the first six months, although both the average return and standard deviation of the investment portfolio show an upward trend, the marginal increase in average return contributes less to the Sharpe ratio than the marginal increase in standard deviation. After the sixth month, as returns and standard deviation stabilize, the Sharpe ratio of hedge funds also tends to stabilize.

Panel (d) further elaborates that as the holding period extends, the *t*-test values of investment returns continuously decline, indicating that the average returns that hedge funds can generate become less significant with increasing holding periods. This also reflects the observation from Panel (a) that while the average return of hedge funds appears to increase with holding period, it is not statistically significant.

Moreover, Fig. 6 distinguishes itself by analyzing returns across three distinct sample sets: those characterized by trend-based characteristics, a combination of trend-based characteristics and risk-based characteristics, and an amalgamation of trend-based, risk-based and macro-derivative characteristics. Despite the variation in characteristics, a consistent pattern emerges across all samples, affirming the universal applicability of the observed trends.

In summary, Fig. 6 provides a nuanced view into how holding period lengths affect the risk-return profile of long-short investment portfolios. Although longer holding periods within six months may offer the temptation of higher returns, their significance in statistical tests is continuously decreasing. This comprehensive analysis not only highlights the strategic considerations investors must weigh but also showcases the robustness of these findings across different investment characteristic samples.

4.3. The performance in different category of hedge funds

In the complex and dynamic world of hedge fund investments, the ability to accurately predict and strategically invest in high-performing funds is paramount. Table 3 delves into this challenge by showcasing the results of an innovative approach using the TFT model to forecast the performance of hedge funds across nine varied categories, specifically focusing on the 10th percentile of funds. This analysis offers insightful revelations about the model's effectiveness in risk mitigation and its capacity to capture lucrative returns across different investment strategies.

Table 3
The performance of fund decile sorted by predicted returns in each category.

	Convertible Arbitrage	Emerging Markets	Equity Market Neutral	Event Driven	Fixed Income Arbitrage	FOFs	Global Macro	Long/Short Equity Hedge	Multi − Strategy
L	0.15	0.07	0.21	0.14	0.11	−0.22	−0.13	−0.01	0.14
	(0.97)	(−0.38)	(1.99)	(1.02)	(1.43)	(−2.14)	(−0.78)	(−0.04)	(1.38)
2	0.19	−0.26	0.13	0.07	0.05	−0.10	−0.17	0.02	0.08
	(1.20)	(−1.60)	(1.26)	(0.56)	(0.50)	(−1.05)	(−1.30)	(0.19)	(0.89)
3	0.18	0.10	0.19	0.20	0.14	−0.00	0.06	0.15	0.13
	(1.16)	(0.61)	(2.81)	(1.74)	(1.44)	(−0.03)	(0.59)	(1.30)	(1.56)
4	0.27	0.13	0.06	0.17	0.22	0.04	0.11	0.19	0.20
	(2.20)	(0.81)	(0.76)	(1.50)	(2.28)	(0.41)	(1.18)	(1.67)	(2.40)
5	0.41	0.36	0.21	0.19	0.15	0.03	0.16	0.25	0.23
	(3.62)	(2.32)	(2.61)	(1.69)	(2.19)	(0.32)	(1.73)	(2.27)	(2.34)
6	0.41	0.25	0.17	0.35	0.23	0.06	0.28	0.23	0.48
	(3.58)	(1.59)	(2.12)	(3.22)	(2.44)	(0.60)	(2.77)	(2.10)	(4.86)
7	0.37	0.18	0.24	0.32	0.37	0.13	0.28	0.31	0.41
	(3.03)	(1.15)	(3.63)	(2.69)	(5.70)	(1.35)	(3.14)	(2.77)	(6.62)
8	0.32	0.39	0.32	0.32	0.53	0.12	0.38	0.42	0.55
	(2.41)	(2.44)	(4.55)	(2.54)	(8.14)	(1.23)	(4.19)	(3.59)	(9.54)
9	0.39	0.42	0.30	0.51	0.65	0.18	0.34	0.52	0.63
	(2.62)	(2.61)	(3.27)	(3.41)	(9.75)	(1.78)	(3.46)	(3.86)	(6.63)
H	0.68	0.44	0.67	0.51	0.76	0.46	0.54	0.54	0.63
	(4.87)	(2.45)	(3.92)	(2.68)	(6.54)	(4.88)	(2.91)	(3.51)	(3.25)
H − L	0.54	0.51	0.48	0.65	0.65	0.24	0.42	0.49	0.57
	(3.70)	(2.75)	(6.60)	(3.85)	(8.70)	(2.05)	(3.22)	(3.23)	(4.92)

This table reports the result of fund deciles sorted by TFT − based predicted returns in 9 categories. In month t, we predict fund returns for month t + 1 for each model and sort funds into deciles based on each model's forecasts. We rebalance it monthly. The predicted top and bottom decile is named as H and L. We report the average monthly return of each strategy, and present its t − value in parentheses below the average monthly return.

The investigation further differentiates the model's predictive accuracy across the diverse landscape of hedge fund strategies. It becomes evident that the TFT model exhibits a particularly enhanced predictive edge for funds focusing on Fixed Income Arbitrage, Convertible Arbitrage and Equity Market Neutral. This distinction is critical, highlighting the model's aptitude in not just identifying safer investment avenues but also in pinpointing where the highest returns can be realized.

Among these strategies, the long-short portfolio of Funds of Funds Hedge Funds produces the lowest average monthly returns. This is due to the fact that Funds of Funds Hedge Funds provide returns that are the average of the returns achieved by other strategies and have the characteristic of double charging. Ultimately, Fixed Income Arbitrage hedge funds stand out, with the highest decile achieving an average monthly return of 0.76 %. Through meticulous analysis and strategic application, the TFT model offers a sophisticated lens through which investors can navigate the intricate hedge fund landscape, optimizing for both risk management and return maximization.

4.4. Characteristic importance

To evaluate the impact of different types of characteristics on the ability to predict fund performance, we conduct an in-depth analysis using Temporal Fusion Transformer (TFT) models equipped with various subsets of predictors. These predictors are categorized into three main groups: trend-based characteristics (1–7 as listed in Table 1), risk-based characteristics (8–14) and macro-derivative features (15–17). Our analysis aims to discern which combination of these characteristics would most accurately forecast fund returns. Our core research finding is that trend-based characteristics provide a strong foundation for predicting hedge fund performance, and incorporating risk-based characteristics and macro-derivative features further improves the stability of our predictions.

Fig. 7 visually represents this finding by depicting the cumulative returns of fund deciles under four different modeling scenarios: using only trend-based characteristics (Panel (a)), combining trend-based characteristics with macro-derivative features (Panel (b)), employing only risk-based characteristics (Panel (c)), and merging risk-based characteristics with macro-derivative features (Panel (d)). In each scenario, returns within the deciles are equal-weighted to ensure a fair comparison.

The results clearly indicate that the most effective model for predicting fund returns is built upon trend-based characteristics, and the inclusion of macro-derivative features within trend-based characteristics does not impact the model's predictions for the top-performing group. In contrast, models based on risk characteristics show weaker predictive capabilities, and when macro-derivative features are added to risk-based models, the ability to distinguish between high-performing and low-performing hedge funds further declines.

Additionally, it is worth noting that during the 2008 financial crisis, when we use trend-based characteristics as our predictive model, we are able to achieve a smaller drawdown rate compared to using risk-based characteristics as our predictive model. This indirectly reflects that during the financial crisis, hedge funds still exhibited their momentum effect, and our model was practical during the crisis period.

Given the concentration of predictability in the extreme deciles, we introduce a long-short prediction portfolio strategy focusing on

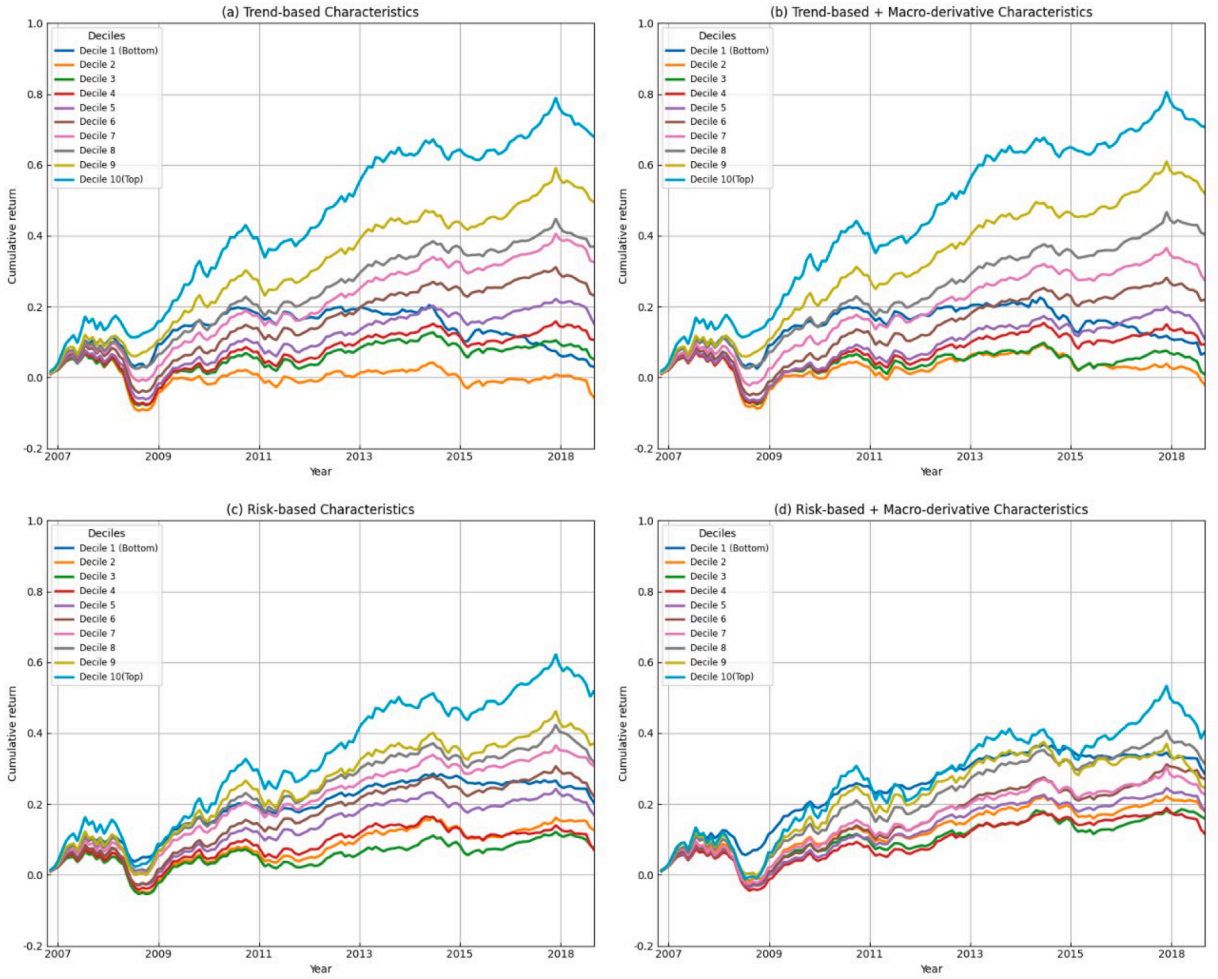


Fig. 7. Cumulative returns of specific factors. These figures show the cumulative returns sorted into prediction deciles for different information sets. The returns are equal-weighted within deciles. We consider trend-based characteristics, trend-based + macro-derivative characteristics, risk-based characteristics, or risk-based + macro-derivative characteristics to predict returns.

the highest and lowest deciles to gauge the skill spread. This approach serves as a practical measure for evaluating the disparity in fund management skill, rather than a direct investment strategy. Fig. 8 illustrates the performance of portfolios that go long on the highest decile of predicted best-performing funds and short on the lowest decile of predicted worst-performing funds, across different information categories.

The outcomes vary significantly depending on the type of information utilized: Portfolios based solely on risk information yield a cumulative out-of-sample return of 31.36 %. Adding macro-derivative features to risk information leads to a return of 11.80 %. Utilizing trend information alone results in a return of 64.91 %. Building on the trend information and adding risk information and macro information respectively leads to cumulative returns of 52.37 % and 63.84 %. The highest return, 85.06 %, is achieved when integrating trend, risk, and macro information. We find an interesting phenomenon that when we individually add risk information and macro information to trend information, it leads to a decrease in the model's predictive power. However, when we incorporate both types of information into trend information simultaneously, the model's predictive power actually increases. These results highlight the substantial impact of incorporating trend and risk analysis into the prediction process, suggesting that a nuanced approach that leverages a comprehensive set of predictors can significantly enhance portfolio performance.

To understand the risk levels of different long-short investment strategies, we examine the Sharpe Ratio for each portfolio strategy, as shown in Table 4. The Sharpe Ratio helps us understand how much return we get for the risk we take. Empirical results indicate that when we add macro information to trend information, there is almost no impact on the model's predictive power, with a monthly average return and Sharpe ratio of 0.45 % and 0.16, respectively. If we do not use trend information as feature variables to train the model, we would not even be able to achieve a positive Sharpe ratio, in other words, the returns we obtain would not exceed the risk-free rate of return. Adding only risk information to trend information would decrease the monthly average return and Sharpe ratio to 0.37 % and 0.1, respectively. However, using the full set of information yields the highest monthly average return and Sharpe ratio, which are 0.57 % and 0.25, respectively. This suggests that risk-like characteristics can enhance the stability of returns obtained using

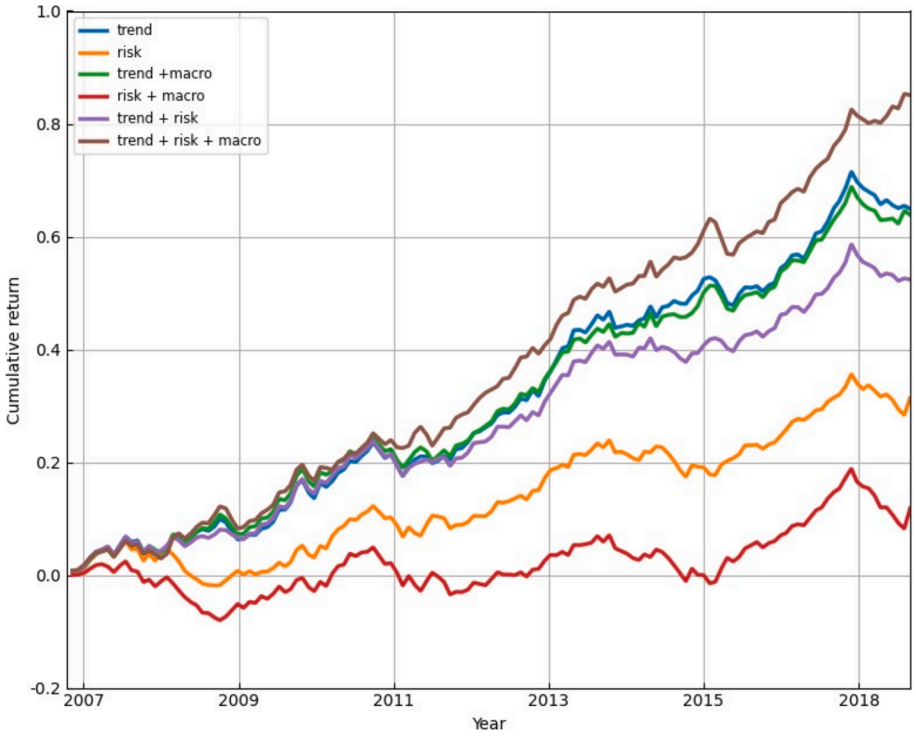


Fig. 8. Cumulative returns of long-short decile portfolios. This figure plots the cumulative returns of equal-weighted long-short decile portfolios that use different information sets for prediction. We consider the combination of trend-based, risk-based characteristics and macro-derivative features.

Table 4
Performance of long-short return portfolios for different information sets.

Information set	Mean(%)	t	SR
Trend	0.45	4.33	0.16
Risk	0.22	2.44	−0.03
Trend + Macro	0.45	4.37	0.16
Risk + Macro	0.08	0.86	−0.15
Trend + Risk	0.37	3.87	0.10
Trend + Risk + Macro	0.57	5.28	0.25

This table reports the Sharpe ratio and mean of equal-weighted long-short decile portfolios that use different information sets for the prediction. We consider six different information sets which combine trend-based, risk-based and Macro-derivative characteristics.

the TFT model.

Furthermore, we calculate the importance of each factor by referencing the method of permutation importance. When calculating permutation importance, we replace the mean squared error (MSE) with the cumulative returns of the long-short investment portfolio, and the final results are shown in Table 5. The results show that the importance of trend-based Characteristics accounts for a total of 59.35 %, with factor R6 having the highest importance of 14.69 %, making it the most important variable among all variables. This may explain why the cumulative returns of the long-short investment portfolio based solely on trend-based Characteristics are much higher than those based solely on risk-based Characteristics, even though the importance of risk-based Characteristics account for a total of 21.47 %. The importance of Macro-derivative Features accounts for a total of 19.18 %, the lowest among the three groups of variables, indicating that while hedge funds also consider Macro-derivative Features, they are not the main determining factor.

When applying machine learning in the hedge fund field, the interpretability of the model is of utmost importance. It is the key to our in − depth understanding of the basis for investment decisions. Through the method of permutation importance, we have gained invaluable insights: in the process of machine − learning − based prediction of hedge funds, trend − based characteristics, especially R6, play the most central role. This result is highly interpretable and is not an obscure black − box output. Past returns, as an intuitive and easily understandable feature, directly reflect the past performance trends of the fund. We can clearly see that it plays a dominant role in the prediction model, which means that funds with excellent past performance are more likely to perform well in the future based on the continuation of this trend, as predicted by the machine − learning model; and vice versa. This interpretable discovery

Table 5
The characteristic importance in prediction.

Variable	Top	Bottom	Permutation importance
R6	67.04	1.11	14.69 %
R9	68.53	3.65	11.19 %
Volatility Beta	69.21	4.07	10.24 %
R1	64.66	9.05	9.86 %
ACF1	67.97	5.78	9.83 %
R3	70.54	4.04	9.11 %
Skew	70.03	4.86	8.84 %
Economic-uncertainty Beta	69.71	8.06	6.34 %
Upside Potential	71.7	7.99	4.67 %
CFVAR	76.64	12.78	3.78 %
CFES	73.36	7.75	3.44 %
Liquidity Beta	74.01	8.07	2.60 %
R12	73.87	8.52	2.33 %
NPES	73.68	9.77	1.41 %
SDI	80.44	3.56	0.93 %
NPVAR	76.62	9.18	0.63 %
Kurt	72.93	12.01	0.11 %

This table shows the cumulative returns of the highest and lowest groups predicted by the TFT model after randomly assigning different variables, as well as the permutation importance calculated based on this. The calculation method for permutation importance is as follows: $\mu_i = \frac{|r - r_i|}{\sum_{i=1}^{n} |r - r_i|}$.
where μ_i represents the permutation importance of the ith factor, r represents the cumulative returns of the long-short investment portfolio before randomly assigning values to each variable, r_i represents the cumulative returns of the long-short investment portfolio after randomly assigning values to the ith factor, and $\sum_{i=1}^{n} |r - r_i|$ represents the sum of the decrease in cumulative returns after randomly assigning values to different factors.

Table 6
Abnormal returns of every category of hedge funds.

Product	Full sample		After 2008		Before 2008	
	α(%)	t	α(%)	t	α(%)	t
Convertible Arbitrage	0.53	42.19	0.29	11.40	0.63	44.94
Emerging Markets	0.47	34.95	0.01	0.76	1.02	49.71
Equity Market Neutral	0.45	40.68	0.23	12.74	0.61	44.06
Event Driven	0.59	63.12	0.20	10.80	0.78	74.51
Fixed Income Arbitrage	0.55	52.04	0.32	17.27	0.68	52.56
Fund of Funds	0.24	77.25	−0.10	−22.68	0.56	133.05
Global Macro	0.45	31.90	0.28	15.28	0.62	28.57
Long/Short Equity Hedge	0.53	87.45	0.19	20.06	0.78	98.41
Multi-Strategy	0.50	73.82	0.35	40.35	0.74	66.16

This table shows the monthly abnormal returns and t-tests of hedge funds with different investment strategies. Due to the differences in investment strategies, we mainly refer to the Fung and Hsieh (2004) seven factors when calculating abnormal returns. We also separately calculate the monthly abnormal returns before 2008, and after 2008 (within 2008).

provides a solid logical foundation for us to apply machine learning models in hedge fund investment decisions. It enables us to formulate investment strategies more confidently and rationally based on the clear importance of features, rather than blindly relying on incomprehensible model results.

Through the above two methods of verification, we can draw the following conclusions: when predicting hedge funds, trend-based features can significantly help the prediction ability of hedge funds, and mainly based on their lagged six-period return rate. Adding risk-based features alone or adding macro-derivative features alone will decrease the model's predictive power, but incorporating both can enhance the model's predictive ability.

4.5. The abnormal return in hedge funds

The question of whether hedge funds can achieve abnormal returns has always been a concern. Unlike the findings of Wermers (2000) that the abnormal returns of mutual funds are zero or negative, studies by Ibbotson, Chen, and Zhu (2011), Kosowski et al.

Table 7
Abnormal returns for each group based on TFT prediction results.

Group	α	t
Low	−0.17	−1.66
2	−0.11	−1.25
3	−0.02	−0.27
4	−0.03	−0.32
5	0.01	0.18
6	0.10	1.26
7	0.12	1.47
8	0.20	2.45
9	0.29	3.21
High	0.42	3.50
High-Low	0.59	5.59

This table shows the predicted results based on the TFT model for grouping hedge funds by month and the monthly average abnormal return rates and t-values that can be obtained by investing in prediction-weighted portfolios. When calculating abnormal return rates, we still use the Fung and Hsieh (2004) seven factor regression.

(2007) have shown that hedge funds generally can achieve abnormal returns of 3 % to 5 %. We apply the Fung and Hsieh (2004) seven-factor[6] regression to calculate the abnormal returns of different hedge funds and arrive at similar conclusions. In Table 6, the regression results show that the more significant types of hedge funds can generally achieve monthly abnormal returns ranging from 0.24 % to 0.59 %. Except for Funds of Funds (FOF) strategy hedge funds, the rest of the hedge funds can achieve an average monthly abnormal return of about 0.5 % throughout the entire sample period, and their ability to obtain abnormal returns has significantly declined after the financial crisis. The FOF strategy hedge funds have the lowest abnormal returns during the entire sample period, with only an average monthly abnormal return of 0.24 %, and are the only hedge fund strategy that cannot obtain abnormal returns after the financial crisis. Although Emerging Markets can achieve the highest abnormal return of 1.02 % per month before the financial crisis, its average monthly abnormal return has dropped to 0.01 % after the crisis. This is mainly because after the financial crisis, country risk also becomes a factor that fund managers had to consider.

We then calculate the abnormal return rates for each group of investments according to the TFT model. Since our test set starts from 2007, when making comparisons, we focus solely on the comparison between the abnormal returns of each group and the abnormal returns of various strategies after the financial crisis. Table 7 shows that the highest decile can achieve a significant monthly average abnormal return rate of 0.42 %, which is higher than the abnormal returns of various hedge fund strategies after the financial crisis. Moreover, since our sample includes data from the 2008 financial crisis period, this may result in a lower abnormal return rate. During normal periods, the TFT model should be able to capture a higher abnormal return rate. This precisely demonstrates that the TFT model can help us better select hedge funds to obtain higher returns exceeding the market.

Furthermore, we refer to the approach of Bali, Brown, and Caglayan (2014a) and others to attribute the abnormal returns obtained from the TFT model. The regression equation we constructed is as follows:

$$\begin{aligned} R_{i,t} = \beta_0 &+ \beta_{1,t} SIZE_{i,t} + \beta_{2,t} AGE_{i,t} + \beta_{3,t} MANFEE_{i,t} + \beta_{4,t} INCFEE_{i,t} \\ &+ \beta_{5,t} REDEM_{i,t} + \beta_{6,t} MININVEST_{i,t} + \beta_{7,t} D_LOCKUP_{i,t} \\ &+ \beta_{8,t} D_LEVERAGE_{i,t} + \varepsilon_{i,t} \end{aligned}, \tag{5}$$

where $R_{i,t}$ is the abnormal return of fund i in month t. *SIZE*, *AGE*, *MGNFEE*, *INCFEE*, *REDEM*, *MININVEST*, *D_LOCKUP*, and *D_LEVERAGE* are fund characteristics: *SIZE* refers to the amount of assets managed by the hedge fund on a monthly basis;*AGE* is the number of months since the hedge fund's inception until the time t; *MGNFEE* is the management fee for the hedge fund, typically around 2 % of the investment amount; *INCFEE* is a percentage fee that the hedge fund charges on the profits generated from the investment; *REDEM* is the redemption notice period for the hedge fund, which is the minimum number of days required for the investor to successfully redeem their investment after submitting a redemption request; *MININVEST* is the minimum amount required for an investor to purchase the hedge fund;*D_LOCKUP* indicates the lock-up period of the hedge fund; *D_LEVERAGE* indicates the use of leverage by the hedge fund.

Many scholars have explored the factors influencing the abnormal returns of hedge funds. We further confirm these scholars' conclusions and extend them based on the TFT model for grouping funds. Table 8 shows that the size and minimum investment amount of hedge funds cannot significantly explain the abnormal returns of hedge funds, which is consistent with the conclusions of Gregoriou and Rouah (2002) and Teo, Koh, and Koh (2003). The age of hedge funds is significantly negatively correlated with their abnormal returns, consistent with Howell (2001)'s research. Regarding the impact of incentive fees on hedge fund abnormal returns, Amenc et al. (2003) show that funds with high incentive fees (greater than or equal to 20 %) outperform funds with low incentive fees in all models used. Our conclusion is that incentive fees are positively correlated with the abnormal returns of hedge funds, but the significance is

[6] The data is from https://people.duke.edu/~dah7/HFRFData.htm.

Table 8
The regression results of abnormal returns.

	INTERCEPT	SIZE	MININVEST	MANFEE	INCFEE	REDEM	AGE	D_LOCKUP	D_LEVERAGE
Low	6.87	−0.83	3.11	−1.88	0.47	0.02	−0.36	0.16	0.12
	(3.41)	(−0.51)	(0.31)	(−1.72)	(4.19)	(0.74)	(−20.16)	(1.18)	(2.27)
2	−5.31	−0.71	−1.73	−1.44	0.20	0.06	−0.05	0.01	0.03
	(−3.40)	(−0.68)	(−0.34)	(−1.80)	(3.09)	(3.72)	(−5.85)	(0.08)	(0.99)
3	−2.80	−1.05	0.27	−1.55	0.11	0.02	−0.02	0.07	−0.09
	(−2.12)	(−1.20)	(0.11)	(−2.19)	(2.10)	(1.82)	(−3.32)	(0.93)	(−3.27)
4	−3.30	−0.14	2.20	−0.00	0.06	0.00	−0.02	0.10	−0.18
	(−2.87)	(−0.18)	(0.88)	(−0.01)	(1.20)	(0.47)	(−3.57)	(1.31)	(−6.84)
5	−2.85	0.18	0.92	0.07	0.06	0.03	−0.04	0.08	−0.14
	(−2.52)	(0.21)	(0.43)	(0.12)	(1.21)	(2.87)	(−5.28)	(1.07)	(−5.53)
6	−0.86	−0.95	−1.90	−0.89	0.13	0.02	−0.02	0.09	−0.14
	(−0.76)	(−1.02)	(−0.76)	(−1.42)	(2.48)	(2.16)	(−2.32)	(1.09)	(−5.44)
7	1.92	−2.73	−1.97	−0.42	0.11	−0.00	−0.03	0.11	−0.09
	(1.63)	(−2.74)	(−0.44)	(−0.65)	(2.02)	(−0.40)	(−3.99)	(1.33)	(−3.33)
8	4.47	−3.12	−7.18	−0.06	−0.06	−0.03	−0.01	0.32	−0.08
	(3.68)	(−2.97)	(−0.84)	(−0.10)	(−1.15)	(−2.02)	(−0.95)	(4.02)	(−2.80)
9	7.72	−0.01	−8.45	−2.01	−0.04	0.03	−0.01	0.11	0.15
	(5.73)	(−0.02)	(−0.28)	(−2.93)	(−0.66)	(1.89)	(−1.45)	(1.41)	(4.77)
High	10.64	1.07	−16.76	−1.64	0.18	0.08	−0.12	−0.02	0.60
	(4.68)	(0.78)	(−0.24)	(−1.46)	(1.75)	(4.10)	(−6.09)	(−0.20)	(12.80)

This table shows the average intercept and average slope coefficients along with their t-values for the regression of abnormal returns. The abnormal returns are the abnormal returns of hedge funds between 2007 and 2018. We conduct regressions on each group of abnormal returns, which are grouped based on the predicted results of the TFT model.

higher for the low-return group than for the high-return group. According to Schneeweis, Kazemi, and Martin (2003), the redemption period seems to affect returns, as funds with quarterly lock-up periods for similar strategies outperform funds with monthly lock-up periods. We find that the redemption period has a significantly greater impact on the highest decile than on the lowest decile. Finally, the use of leverage also has a significant positive impact on the returns of hedge funds.

4.6. *The performance in high and low narrative fund attention*

Having discussed how the TFT model can help us capture the abnormal returns from hedge funds and the attribution of abnormal returns, we further explore the abnormal returns that we are able to capture when a hedge fund is in a high-attention period and a low-attention period. We decompose the abnormal returns of the hedge fund into two components: the trading component between high-attention periods (i.e. trades within low-attention periods), and the trading component within high-attention periods.

$$R_{i,t}^{abnormal} = \widehat{R_{i,t}} - f_t\widehat{\beta_i} + R_{i,t} - f_t\beta_i - \left(\widehat{R_{i,t}} - f_t\widehat{\beta_i}\right) \tag{6}$$

The abnormal return between high attention periods refers to the abnormal return of an investor who invests in the fund holdings during the most recent high attention period and holds the investment portfolio until the next high attention period, denoted as $\widehat{R_{i,t}} - f_t\widehat{\beta_i}$ in the formula. In the equation, $\widehat{R_{i,t}}$ represents the assumed return of a hedge fund i, which fixes its investment portfolio weights at the level of the last high attention period (at $t-1$). f_t is the simultaneous return vector on the Fung and Hsieh (2004) seven factors, while $\widehat{\beta_i}$ is the vector of Fung and Hsieh (2004) seven-factors exposures related to the assumed fund returns $\widehat{R_{i,t}}$. $R_{i,t} - f_t\beta_i - \left(\widehat{R_{i,t}} - f_t\widehat{\beta_i}\right)$ represents the abnormal return rate during low attention periods, $R_{i,t} - \widehat{R_{i,t}}$ is the return gap, and $f_t(\beta_i - \widehat{\beta_i})$ is the risk difference.

A relatively high abnormal return during low attention periods indicates that the fund can add value by actively trading when hedge fund attention is low. We argue that there are two main reasons why hedge funds earn lower returns during high-attention periods than during low-attention periods: First, the periods of increased attention by the *Wall Street Journal* to hedge funds tend to be periods when hedge funds are in financial distress, and the average returns available to the hedge fund industry are low, leading to a low average return for the portfolio. Second, during normal times, hedge funds exhibit herding behavior, and when aggregate hedge fund performance is poor, there is a flight from herding that depresses hedge fund returns further.

The abnormal return during low attention periods can be further decomposed into two parts: the return gap as defined by Kacperczyk, Sialm, and Zheng (2008) which represents the abnormal returns achieved as a result of the hedge fund's selection and the risk exposure differential which is the difference between the risk exposure of the hypothetical fixed portfolio during low attention periods and the actual risk exposure of the current (rebalanced) portfolio. As we can see in Table 9, returns during periods of low attention are primarily driven by the return gap, indicating that the TFT model has a strong advantage in hedge fund selection. Additionally, we observe that risk differences have a negative impact on returns, suggesting that the TFT model primarily contributes to abnormal returns through fund selection rather than risk exposure management.

Table 10 shows the results of abnormal returns when we go long on the highest decile of hedge funds and short on the lowest decile based on the predictions from univariate analysis. (Although it is not a feasible measure, we do this for the convenience of analysis.) We

Table 9
Decomposition of univariate long-short abnormal return factors.

Factor	Total		High attention periods		Low attention periods		Risk difference		Return gap	
	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean
R1	0.40	0.46	0.31	0.12	0.33	0.34	−0.13	−0.05	0.35	0.39
R6	0.37	0.44	0.20	0.09	0.33	0.35	−0.15	−0.05	0.36	0.40
R3	0.35	0.43	0.22	0.11	0.31	0.32	−0.17	−0.06	0.34	0.38
R9	0.25	0.25	0.07	0.03	0.28	0.22	−0.19	−0.07	0.34	0.29
R12	0.23	0.21	0.12	0.04	0.23	0.17	−0.16	−0.05	0.28	0.22
NPVAR	0.21	0.16	0.20	0.15	0.05	0.01	−0.16	−0.01	0.10	0.02
CFES	0.18	0.13	0.17	0.11	0.06	0.02	−0.20	−0.02	0.13	0.03
NPES	0.16	0.13	0.16	0.12	0.07	0.01	−0.17	−0.01	0.11	0.02
CFVAR	0.12	0.09	0.11	0.08	0.03	0.01	−0.17	−0.01	0.08	0.02
acf1	0.12	0.11	0.10	0.09	0.10	0.01	−0.10	0.00	0.11	0.02
SDI	0.12	0.10	0.10	0.09	0.09	0.02	−0.17	0.00	0.12	0.02
kurt	0.12	0.11	0.11	0.10	0.05	0.01	−0.11	0.00	0.06	0.01
euncbeta	0.10	0.07	0.09	0.06	0.04	0.01	0.24	0.01	0.00	0.00
vixbeta	0.05	0.03	0.04	0.02	0.03	0.01	−0.16	−0.01	0.07	0.02
illiqbeta	0.04	0.03	−0.02	−0.02	0.24	0.05	−0.05	0.00	0.25	0.05
UpsidePotential	0.03	0.02	0.05	0.04	−0.12	−0.01	−0.05	0.00	−0.10	−0.01
skew	−0.02	−0.01	−0.03	−0.02	0.06	0.01	−0.08	0.00	0.08	0.01

This table reports mean and Sharpe ratio of the decomposition of univariate long-short abnormal return factors. Means of abnormal returns are reported in percentages. The results are sorted according to the Sharpe ratio of the long-short factors. For each of the 17 characteristics and each abnormal return, we construct decile-sorted portfolios. The long-short factors are the differences between the top decile and the bottom decile.

Table 10
Decomposition of prediction long-short return portfolios.

Factor	Total		High attention periods		Low attention periods		Risk difference		Return gap	
	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean
Trend	0.34	0.40	0.16	0.08	0.31	0.32	−0.17	−0.06	0.35	0.38
Risk	0.14	0.11	0.17	0.06	0.09	0.05	−0.16	−0.09	0.18	0.14
Trend + Macro	0.36	0.42	0.25	0.11	0.30	0.31	−0.15	−0.05	0.33	0.36
Risk + Macro	−0.03	−0.03	−0.14	−0.05	0.04	0.02	−0.16	−0.08	0.13	0.10
Trend + Risk	0.30	0.31	0.25	0.10	0.23	0.21	−0.16	−0.06	0.27	0.26
Trend + Risk + Macro	0.49	0.59	0.33	0.17	0.38	0.42	−0.13	−0.03	0.40	0.45

This table reports the mean and Sharpe ratio for the decomposition of the equal-weighted long-short decile portfolios. We use different information sets to predict returns. Means of abnormal returns are reported in percentages.

calculate the total mean and Sharpe Ratio for each individual variable, as well as the means and Sharpe Ratios for the four groups: high attention periods, low attention periods, risk difference, and return gap. Columns 1–5 of Table 9 all represent trend-based characteristics, which indirectly highlight the importance of trend-based characteristics in predicting hedge fund returns.

It is also observed in Table 10 that for most single-factor predictions, the abnormal returns obtained when constructing portfolios are lower during high attention periods compared to abnormal returns during low attention periods. This phenomenon is more pronounced in Table 9, indicating that we can achieve higher returns by actively selecting hedge funds during low attention periods.

5. Economic analysis

5.1. Economic gain from diversification

So far, we have independently studied hedge funds. Subsequently, we attempt to construct a portfolio combining hedge funds and risk-free assets. Since hedge funds do not have a short-selling mechanism, we only select the top-tier hedge funds when choosing them. Based on the prediction results of the TFT model, we consider the economic benefits of mean–variance investors' allocation between risk-free assets and hedge funds. Following Ma, Li, and Zhang (2024), the utility function of the mean–variance preference is as follows:

$$u(W_{t+1}) = E[W_{t+1}] - \frac{\gamma}{2} Var[W_{t+1}], \tag{7}$$

where W_{t+1} represents the investor's wealth at time $t+1$, u is the investor's utility function, and γ denotes the investor's degree of risk aversion. We assume an investor's risk aversion level is 2. Then, the proportion of the investor's allocation to the risky asset (hedge fund) is,

Table 11
Conditional portfolio for mean–variance preference.

Portfolio	Mean(%)	Std(%)	SR	Δ CER(%)
Convertible Arbitrage	0.22	0.91	0.24	0.28
Emerging Markets	0.11	0.98	0.12	0.03
Equity Market Neutral	0.30	1.02	0.29	0.14
Event Driven	0.21	0.97	0.21	0.15
Fixed Income Arbitrage	0.40	0.87	0.46	0.52
Fund of Funds	0.16	0.92	0.17	0.20
Global Macro	0.21	1.04	0.20	0.00
Long/Short Equity Hedge	0.15	0.99	0.15	0.05
Multi-Strategy	0.30	1.00	0.30	0.18
All Hedge Fund	0.23	0.96	0.24	0.19

The table reports the performance of portfolios that include risk-free assets, which are constructed based on the prediction results of hedge funds under various strategies according to the TFT model. We construct portfolios using the top decile hedge funds and risk-free assets, and present the results for nine different strategies for both hedge funds and all hedge funds combined. We assume an investor's risk aversion level to be 2. Investors allocate their assets monthly between the top-tier hedge funds and risk-free assets. Δ CER is the certainty equivalent return (CER) gain that an investor achieves by investing in a portfolio of hedge funds and risk-free assets instead of investing solely in risk-free assets. We also report the mean (Mean), standard deviation (Std) of returns, and Sharpe ratio (SR) of the different portfolios.

$$\alpha_t = \frac{1}{\gamma W_t} \frac{E\left[R_{t+1}^{\text{predict}} - R_{f,t+1}\right]}{Var\left[R_{t+1}^{\text{predict}} - R_{f,t+1}\right]}. \tag{8}$$

The weight of the investor's allocation to the risky asset is directly proportional to the expected excess return at the next moment and inversely proportional to the variance of the expected excess return at the next moment. When calculating the variance of the expected excess return at the next moment, we use a one-year rolling window of past monthly returns for estimation. Since hedge funds do not have a short-selling mechanism, referring to Joenväärä et al. (2021), we restrict the weight of the risky asset to be between 2 % and 60 %. The return of this portfolio at time t + 1 is

$$R_{p,t+1} = (1 - \alpha_t) R_{f,t+1} + \alpha_t R_{t+1}^{\text{hedge fund}}. \tag{9}$$

Subsequently, we construct portfolios based on the prediction results of hedge funds for each type of investment strategy and calculate the Certainty Equivalent Return (CER) and Sharpe Ratio for each portfolio. This helps us understand the economic value of the portfolios. The calculation method for the Certainty Equivalent Return (CER) is as follows:

$$CER_p = \widehat{\mu}_p - \frac{1}{2}\gamma \widehat{\sigma}_p^2, \tag{10}$$

where $\widehat{\mu}_p$ represents the sample average monthly return of the portfolio, and $\widehat{\sigma}_p^2$ represents the variance of the portfolio's monthly returns. The CER is used to measure the risk-free rate of return that an investor is willing to accept when facing a risky investment, so that their utility is equivalent to that of the risky investment. The CER gain is the difference between the CER of an investor who invests in a hedge fund and a risk-free asset portfolio and the CER of an investor who invests only in risk-free assets. It represents the excess utility that an investor obtains from portfolio investment.

Table 11 shows the performance of portfolios that include risk-free assets under various hedge fund strategies and for all hedge funds combined. After adding risk-free assets to hedge funds employing the Fixed Income Arbitrage strategy, the best performance can be achieved. This portfolio has the highest average monthly return and Sharpe ratio among all portfolios, at 0.4 % and 0.46, respectively. The CER gain of this portfolio is 0.52 %, indicating that investing in this portfolio can increase an investor's utility by 0.52 % compared to holding only risk-free assets. Although the CER gains of portfolios constructed based on other strategies are lower, they are all non-negative, which further demonstrates that incorporating hedge funds selected based on the prediction results of the TFT model into risk-free assets can bring higher utility to investors.

Although we have demonstrated that incorporating hedge funds into risk-free assets can provide investors with higher utility, in real life, the vast majority of people still invest in risky assets such as stocks. Therefore, it is necessary to consider the impact of adding hedge funds to a stock portfolio on investors.

Now let's examine whether adding hedge funds to a passive portfolio (R_P) can bring economic benefits to investors. Following the approach of Joenväärä et al. (2021), we assume that a passive portfolio is composed of 60 % stocks and 40 % bonds, where the stocks are represented by the S&P 500 and the bonds by the Vanguard Total Bond Market Index Fund. We construct an active portfolio (R_a) by investing W% in hedge funds and (100-W)% in the passive portfolio, where the weight of hedge funds ranges from 2 % to 60 % and remains constant over time. We rebalance the asset weights monthly to ensure the investment ratio between hedge funds and the passive portfolio is maintained at W:(100-W).

When measuring the economic value to investors of switching from a passive portfolio to an active portfolio, we employ a utility-based approach that represents the maximum fee an investor would be willing to pay for making the switch from the passive to the

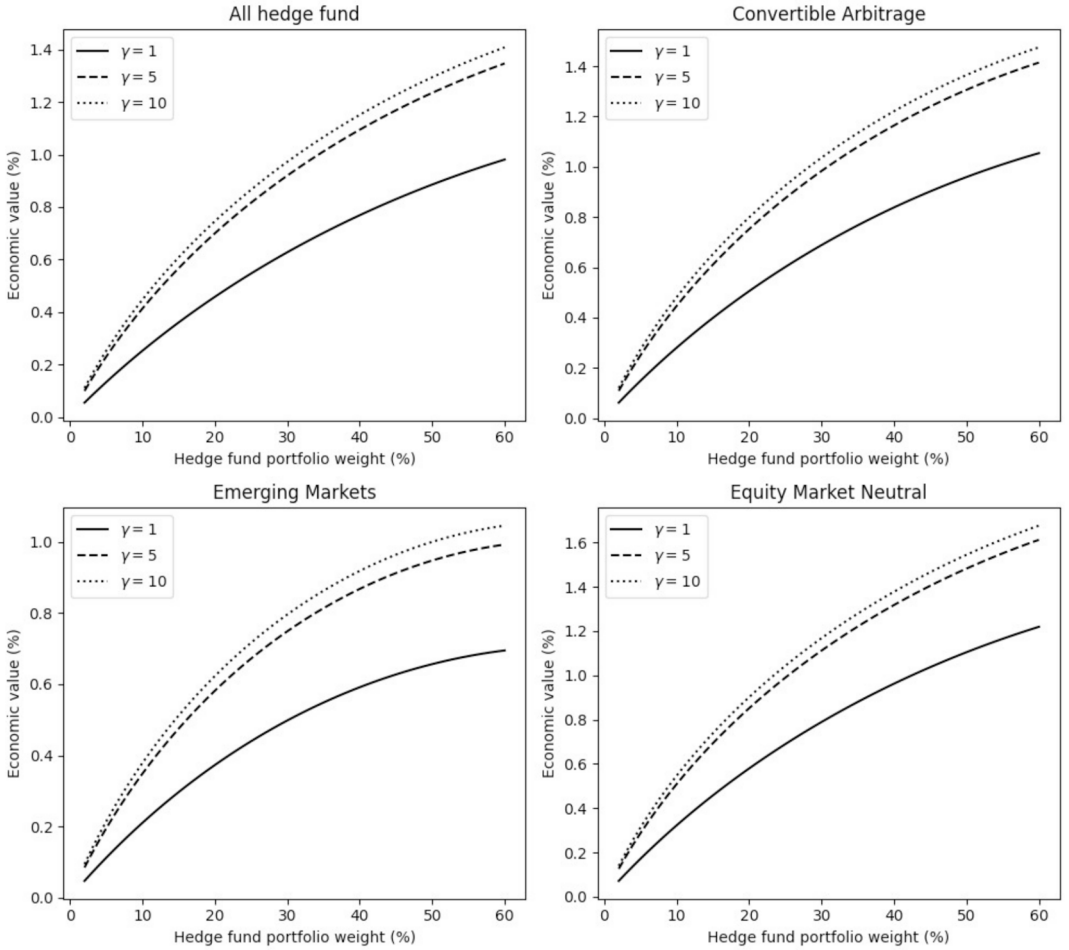


Fig. 9. The economic value of including hedge funds in traditional assets. This figure illustrates the economic benefits from 2007 to 2018 of adding the TFT-based toptier hedge funds to existing passive portfolio. The passive portfolio consists of 60 % stocks (S&P 500 Index) and 40 % bonds (VBTIX). The active portfolio is composed of w% hedge funds and (100-w)% of the passive portfolio, where w% varies from 2 % to 60 %. All portfolios are buy-and-hold portfolios that are rebalanced monthly, with hedge funds being reselected at the end of each month. To measure the economic value in a benchmark-independent manner, we use the Δ measure based on utility, which can be interpreted as the maximum fee that an investor would be willing to pay to switch from the passive portfolio to the active portfolio. We use risk aversion levels of $\gamma = 1, 5, 10$ in the quadratic utility under the Δ measure to accommodate different investor preferences.

active portfolio. The calculation is as follows:

$$U(R) = W_0\left(\sum_{t=1}^{T} R_t - \frac{\gamma}{2(1+\gamma)}R_t^2\right), \tag{11}$$

In the formula, W_0 represents the investor's initial wealth, R_t represents the investor's return at time t, and γ represents the investor's degree of risk aversion. Furthermore, based on

$$U(R_a - \Delta) = U(R_p), \tag{12}$$

the economic value for an investor to switch from a passive portfolio to an active portfolio can be calculated as Δ, and the calculation method is as follows:

$$\eta = \frac{\gamma}{1+\gamma}, \tag{13}$$

$$Y = E[R_{p,t}] - E[R_{a,t}] - \frac{\eta}{2}\left(E\left[R_{p,t}^2\right] - E\left[R_{a,t}^2\right]\right), \tag{14}$$

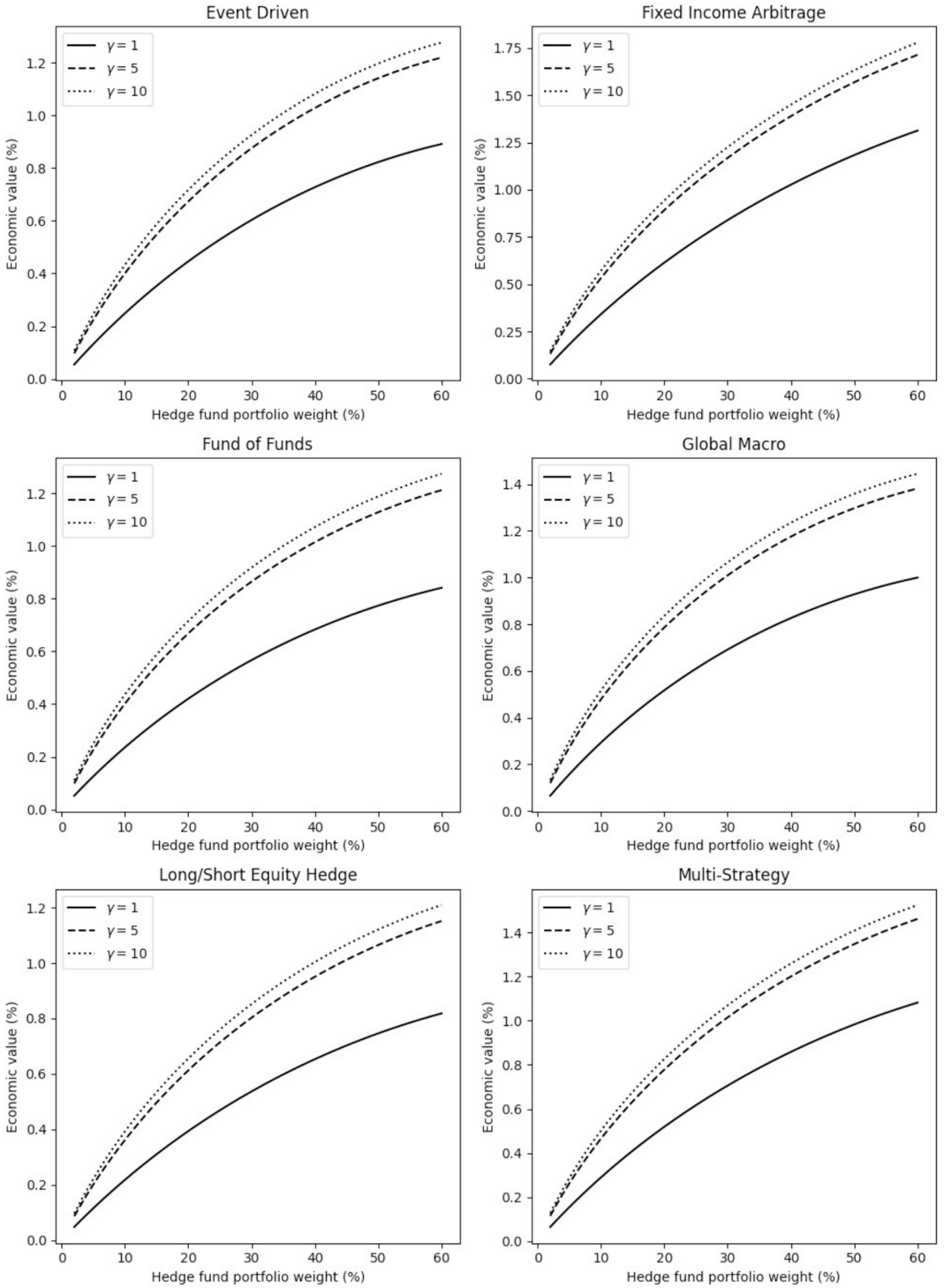


Fig. 9. (*continued*).

Table 12
The portfolio performance considering different macroeconomic statement.

Factor	Total		High value periods		Low value periods		Risk difference		Return gap	
	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean
DIV	0.49	0.59	0.37	0.37	0.24	0.22	0.13	0.02	0.22	0.20
INF	0.49	0.59	0.41	0.51	0.13	0.08	0.14	0.02	0.10	0.07
UNEMP	0.49	0.59	0.42	0.41	0.16	0.19	0.19	0.06	0.10	0.13
MKT	0.49	0.59	0.38	0.41	0.25	0.18	0.12	0.02	0.22	0.16
GDP	0.49	0.59	0.46	0.54	0.09	0.05	−0.01	0.00	0.09	0.05
RREL	0.49	0.59	0.44	0.50	0.17	0.09	0.06	0.03	0.10	0.07
TERM	0.49	0.59	0.40	0.43	0.21	0.16	0.01	0.00	0.21	0.16

This table reports the mean and Sharpe ratio for the equal-weighted long-short decile portfolios in the high and low levels of macroeconomic state. We consider seven macro indicators including DIV, GDP, INF, MKT, RREL, TERM, and UNEMP (Bali et al. 2014b), defining periods above (below) the mean as high (low) value periods. DIV is the dividend yield of the S&P 500 Index, GDP is the monthly growth rate of real per capita GDP in the United States, INF is the inflation rate calculated based on the U.S. CPI, MKT is the excess return of the U.S. equity market index, RREL is the difference between the three-month T-bill rate and its 12-month backward moving average, TERM is the difference between the yields on ten-year and three-month Treasury securities in the United States, and UNEMP is the monthly unemployment rate in the United States. Means of abnormal returns are reported in percentages.

$$D = \left(1 - \eta E\left[R_{a,t}\right]\right)^2 - 2\eta Y, \tag{15}$$

$$\Delta = \frac{\sqrt{D} + \eta E\left[R_{a,t}\right] - 1}{\eta}. \tag{16}$$

Fig. 9 illustrates the economic value of hedge funds under different investment strategies. They share a common characteristic: as the proportion of investment in hedge funds increases, their economic value also rises. Moreover, investing in hedge funds can bring higher economic value to investors with a higher degree of risk aversion, indirectly indicating that investing in hedge funds can bring greater utility to traditional investors, and even more so for those with a higher risk aversion. With a 60 % allocation to hedge funds, investors with a risk aversion level of 10 can achieve an economic return of 1.2 % to 1.75 % from investing in hedge funds. Among them, investing in hedge funds using the Fixed Income Arbitrage strategy can bring the highest economic return of 1.75 % to investors.

5.2. Macro-timing ability of TFT

Finally, we adopt the method of equation (6) to examine the macro-timing ability of TFT by considering seven macroeconomic indicators, including DIV, GDP, INF, MKT, RREL, TERM, and UNEMP (Bali, Brown, and Caglayan, 2014b). DIV is the dividend yield of the S&P 500 Index, GDP is the monthly growth rate of real per capita GDP in the United States, INF is the inflation rate calculated based on the U.S. CPI, MKT is the excess return of the U.S. equity market index, RREL is the difference between the three-month T-bill rate and its 12-month backward moving average, TERM is the difference between the yields on ten-year and three-month Treasury securities in the United States, and UNEMP is the monthly unemployment rate in the United States. For data updated quarterly, we used linear interpolation to fill in the missing months.

Table 12 shows the results. The TFT-based portfolios achieve higher returns when the macroeconomy is relatively good. This phenomenon is most evident when using GDP as a macroeconomic indicator, with the monthly average returns and Sharpe ratios during high (low) value periods being 0.54 % (0.05 %) and 0.46 (0.09), respectively. The finding concurs with the stable performance of TFT model in different macroeconomic conditions.

6. Conclusion

In this paper, we apply the generative AI model to the study on hedge funds for the first time. We find our method can successfully differentiate high from low-performing funds across variance investment strategies. Trend-based characteristics dominate the prediction of hedge fund returns. Abnormal returns among hedge funds are primarily related to fund age, management fees, and incentive fees. In terms of narrative fund attention, hedge fund abnormal returns mainly stem from active selections during periods of low media attention. Diversified investment that allocates hedge funds within traditional assets can bring higher utility to investors. This study highlights the broader implications of fund selection and allocation in the era of big data by exploring machine learning's predictive power from a high-dimensional perspective. The insights from this study hold significant potential for future fund-level investigations.

CRediT authorship contribution statement

Tian Ma: Supervision, Funding acquisition, Data curation, Conceptualization. **Wanwan Wang:** Writing – original draft, Visualization, Validation, Software, Methodology. **Fuwei Jiang:** Writing – review & editing, Methodology, Funding acquisition, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The variable construction

1. R1, R3, R6, R9, R12: Our trend-based features are calculated as the accumulated returns in past (1, 3, 6, 9, 12) months.
2. ACF1: First-order autocorrelations of hedge fund returns. We calculate the correlation between fund return (t:t-35) with the lagged return (t-1:t-36) as ACF1.
3. UP: Upside Potential. We calculate the max monthly return in past 36 month as the UP.
4. NPVAR: Value at Risk at empirical distribution. We calculate the quantile of the past 36-month fund returns as NPVAR.
5. CFVAR: Value at Risk at Cornish-Fisher distribution. Following Cornish and Fisher (1973), we calculate the CFVAR as:

$$CFVAR_t = Mean(R_{t:t-35}) - w^{*}Std(R_{t:t-35})$$

$$w = z + \frac{1}{6}(z^2-1)s + \frac{1}{24}(z^3-3z)k - \frac{1}{36}(2z^3-5z)s^2$$

z = 1.645, s is the skewness of the past return, k is the kurtosis of the past return.

6. NPES: Expected Shortfall at empirical distribution. We calculate mean monthly return below the NPVAR as NPES.
7. CFES: Expected Shortfall at Cornish-Fisher distribution. We calculate mean monthly return below the CFVAR as CFES.
8. SDI: Distinctiveness of a fund's investment strategy. We calculate the correlation between fund return and the monthly mean strategy return (such as: convertible arbitrage, dedicated short bias et al.) it belongs to in past 36 month.

$$DLME_t = \frac{LTL_t}{MV_t}$$

9. Skew and Kurt: we calculate the skewness and kurtosis of the past 12 months returns as Skew and Kurt.
10. Illiqbeta: Liquidity Beta. We run the following regression in past 36 months to calculate the Illiqbeta. Where MKT is the fund market return and ILLIQ is the liquidity factor following Pastor and Stambaugh (2003).[7]

$$R_{i,t} = \beta_1 MKT + \beta_{ill} ILLIQ + \mu_{i,t}$$

11. Euncbeta: Economic uncertainty beta. We run the following regression in past 36 month to calculate the Illiqbeta. Where MKT is the fund market return and EUNC is the Economic Uncertainty index.[8]

$$R_{i,t} = \beta_1 MKT + \beta_{\text{eun}} EUNC + \mu_{i,t}$$

12 Vixbeta: Vix Beta. We run the following regression in past 36 month to calculate.where MKT is the fund market return and VIX is the Volatility Index.

$$R_{i,t} = \beta_1 MKT + \beta_{VIX} VIX + \mu_{i,t}$$

Appendix B. . Individual components of TFT model

2.1. Gated residual network

The exact relationship between exogenous inputs and targets is often unknown a priori, making it challenging to predict which variables are relevant. Moreover, determining the appropriate extent of nonlinear processing is difficult, and simpler models may

[7] See: Pastor, L., & Stambaugh, R. F. (2003). Liquidity risk and expected stock returns. Journal of Political economy, 111(3), 642–685.
[8] See: Bali, T. G., Brown, S. J., & Caglayan, M. O. (2014). Macroeconomic risk and hedge fund returns. Journal of Financial Economics, 114(1), 1–19.

suffice in certain situations—for instance, when the dataset is small or noisy. To enable the model to apply nonlinear processing flexibly only, when necessary, the Temporal Fusion Transformer (TFT) introduces the Gated Residual Network (GRN).

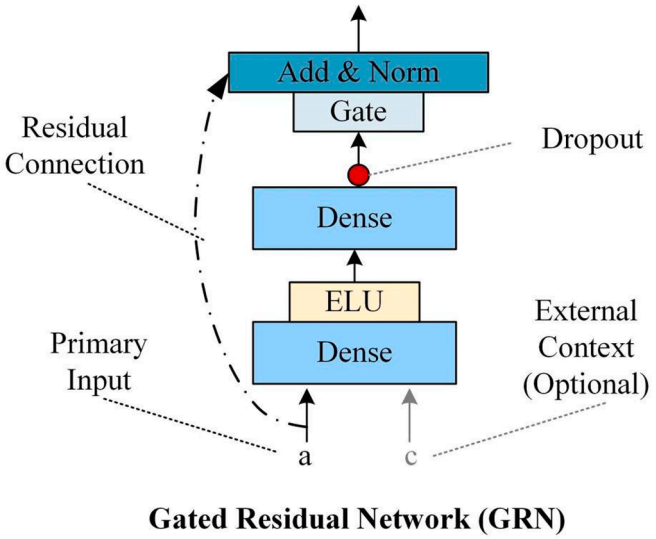


Fig. A1. Gated Residual Network. This figure shows the encoder structure, which contains three building blocks, which are multi-headed attention modules, two kernel sizes 1 in front of the convolution, and two residual connections after each module.

The GRN takes a primary input and an optional context vector, with its computation formula as follows:

$$GRN\omega(a,c) = LayerNorm(a + GLU\omega(\eta_1)) \tag{17}$$

$$\eta_1 = W_{1,\omega}\eta_2 + b_{1,\omega} \tag{18}$$

$$\eta_2 = ELU\left(aW_{2,\omega} + cW_{3,\omega} + b_{2,\omega}\right) \tag{19}$$

where ELU denotes the Exponential Linear Unit activation function, represents an intermediate layer, LayerNorm refers to standard layer normalization, and indicates how weights are shared across the network. The GLU (Gated Linear Unit) is employed to suppress unnecessary components within a given dataset and provides adaptive depth and network complexity to accommodate a wide range of datasets and scenarios. Letting R, the computation for GLU is as follows:

$$GLU_{\omega}(\gamma) = \sigma\left(W_{3,\omega}\gamma + b_{3,\omega}\right) \odot \left(W_{4,\omega}\gamma + b_{4,\omega}\right) \tag{20}$$

where σ is the sigmoid activation function, W and b are weights and biases, $\odot$ represents the Hadamard product (element-wise multiplication), and d is the size of the hidden state.

2.2. Static information encoders layer

In contrast to earlier models for forecasting time series, the TFT integrates data from static inputs, utilizing distinct Gated Residual Network (GRN) encoders to produce four types of vectors. These vectors serve unique functions within the temporal fusion decoder module of the TFT. The role of static information is crucial in several aspects: (1) selecting temporal variables; (2) local processing of temporal features; (3) enhancing temporal features. For instance, considering the vector for temporal variable selection (c_s) and the output of gated residual network(ζ), its purpose is outlined as follows.

$$c_s = GRN_{c_s}(\zeta) \tag{21}$$

2.3. Multi-head attention mechanism

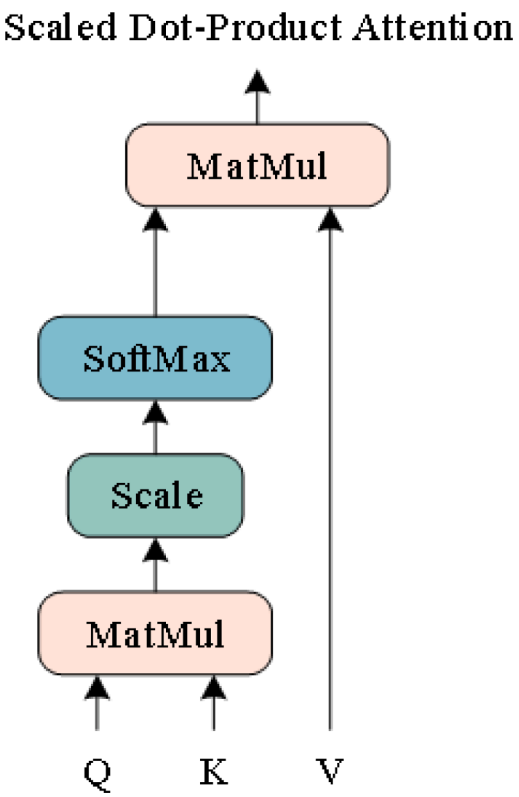


Fig. A2. Multi-head attention. This figure illustrates the computational process of scaled dot-product attention.

The crux of multi-head attention lies in the use of multiple independent attention calculations, which function together as an integrated entity to mitigate over fitting. This approach involves conducting multiple queries in parallel to extract multiple pieces of information from the input data. Each attention module focuses on a distinct aspect of the input data. In each attention module, a sequence entry called 'query' (Q) is compared to all other sequence entries, known as 'keys' (K), through a scaled dot product, with the query and key embedding dimensions scaled equally. The resulting output is then used to weight the same sequence entries, called 'values' (V). Prior to the scaled dot product attention, Q, K, and V are first subjected to a linear transformation. Importantly, there are h calculations, with each calculation counted as a single head. It is referred to as multi-head attention, and the parameters W for each linear transformation of Q, K, and V are distinct. The scaled dot product attention outcomes for h calculations are subsequently concatenated, and the linear transformation results are used as the output of the multi-headed attention. The primary advantage of multi-headed attention is its ability to expand the attention model's capacity to attend to information at various points in time, providing multiple 'representational subspaces' at the attention layer. The multi-head attention function is presented as follows:

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{22}$$

The multi-head attention function is as follows:

$$MultiHead(Q,K,V) = Concat(head_1,\cdots,head_n)W^O \tag{23}$$

where

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{24}$$

where $W_i^Q \in R^{d_{model}\times d_Q}$, $W_i^K \in R^{d_{model}\times d_K}$, $W_i^V \in R^{d_{model}\times d_V}$, $W_i^O \in R^{d_{model}\times hd_v}$ are parameter matrices. In this work, we employ $h = 8$. parallel attention layers, or heads. For each of these, we use $d_v = d_k = d_{model} = 64$.

2.4. Temporal fusion decoder

To capture the temporal relevance within the dataset, the temporal fusion decoder is meticulously constructed with several layers, culminating in a final layer responsible for generating the output. The notation $\tau \in \{1,\cdots,\tau_{max}\}$ denotes the prediction step.

(I) Layer for Processing Local Correlations:

In time series analysis, the significance of a data point is often intimately linked with its neighboring values. This research adopt a sequence-to-sequence approach to grasp this local dependency. For the initial layer of the temporal fusion decoder, a gated skip connection strategy is implemented.

The equation for this process is given by

$$\widetilde{\phi}(t,n) = LayerNorm\left(\widetilde{\xi_{t+n}} + GLU_{\phi}(\phi(t,n))\right) \tag{25}$$

(II) Layer for Encoding Static Information:

Given that static information can significantly impact the accuracy of time series forecasts, this study introduce a layer specifically for encoding static input variables. This static enrichment layer is defined according to Equation (26), where n represents the position index, and c_e is the context vector.

$$\theta(t,n) = GRN_{\theta}\left(\widetilde{\phi}(t,n), c_e\right) \tag{26}$$

(III) Layer for core self-attention:

The subsequent layer incorporates self-attention mechanisms, focusing on the forecast time t. Here, $H(t)$ represents the matrix for static-enriched temporal features. $B(t)$ is defined as the result of the Interpretable Multi-Head operation on the matrix $H(t)$. To enhance the training process, an additional gating layer is introduced after the temporal self-attention layer.

$$B(t) = InterpretableMultiHead(H(t), H(t), H(t)) \tag{27}$$

$$\delta(t,n) = LayerNorm(\theta(t,n) + GLU_{\delta}(\beta(t,n))) \tag{28}$$

(IV) Forward propagation layer

The fourth layer processes the computational outcomes from the self-attention architecture. Additionally, within the Temporal Fusion Transformer (TFT), a Gated Residual Network (GRN) is integrated, facilitating a direct connection to the subsequent layer.

$$\psi(t,n) = GRN_{\psi}(\delta(t,n)) \tag{29}$$

$$\widetilde{\psi}(t,n) = \left(\widetilde{\phi}(t,n) + GRN_{\widetilde{\psi}}(\psi(t,n))\right) \tag{30}$$

$$\widehat{y} = W\widehat{\psi}(t,\tau) + b \tag{31}$$

Table A1
Selection of hyperparameters for machine learning.

Model	Hyperparameters	Candidates	Optimal
Temporal Fusion Transformer Model	Attention head size	[1,64]	8
	Hidden continuous size	[1,64]	8
	Hidden size	[1,64]	8
	Dropout	0.01,0.05,0.1,0.8	0.01
	Learning rate	0.01,0.005,0.001,0.0001	0.01

This table describes our hyperparameter settings for machine learning method, where the parameters in the attention brackets are the number of heads, values, keys, and d_{model} in that order.

Data availability

Data will be made available on request.

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